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KINEMATICS OF ROTATION MOTION

Rigid Body

A rigid body is an assemblage of a large number of material particles, which do not change their mutual distances under any circumstance or in other words, they are not deformed under any circumstance.

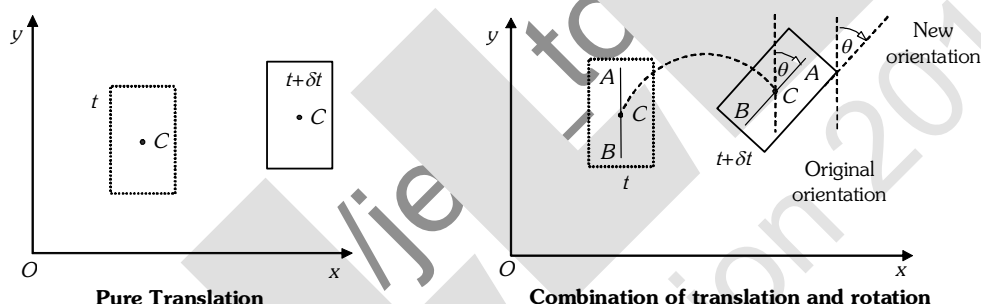
Actual material bodies are never perfectly rigid and are deformed under action of external forces. When these deformations are small enough to be considered during their course of motion, the body is assumed a rigid body. Hence, all solid objects such as stone, ball, vehicles etc are considered as rigid bodies while analyzing their translation as well as rotation motion.

To analyze rotation of a body relative motion between its particles cannot be neglected and size of the body becomes a considerable factor. This is why study of rotation motion is also known as *mechanics of rigid bodies*.

Rotation Motion of a Rigid Body

Any kind of motion of a body is identified by change in position or change in orientation or change in both. If a body changes its orientation during its motion it said to be in rotation motion.

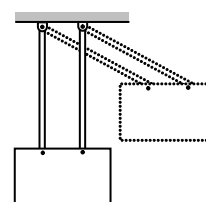
In the following figures, a rectangular plate is shown moving in the x - y plane. The point C is its mass center. In the first case it does not changes orientation, therefore is in pure translation motion. In the second case it changes its orientation by during its motion. It is a combination of translation and rotation motion.



Rotation i.e. change in orientation is identified by the angle through which a linear dimension or a straight line drawn on the body turns. In the figure this angle is shown by θ .

Ex. Identify Translation and rotation motion

A rectangular plate is suspended from the ceiling by two parallel rods each pivoted at one end on the plate and at the other end on the ceiling. The plate is given a side-push to oscillate in the vertical plane containing the plate. Identify motion of the plate and the rods.



Sol.

Neither of the linear dimensions of the plate turns during the motion. Therefore, the plate does not change its orientation. Here edges of the body easily fulfill our purpose to measure orientation; therefore, no line is drawn on it.

The plate is in *curvilinear translation motion* and the rods are in rotation motion.

Types of Motions involving Rotation

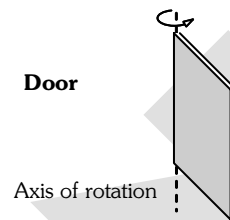
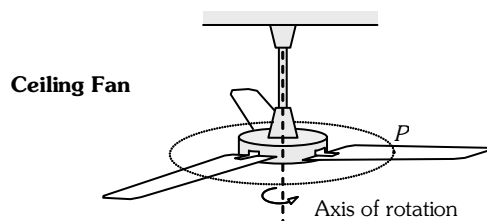
Motion of body involving rotation can be classified into following three categories.

- I Rotation about a fixed axis.
- II Rotation about an axis in translation.
- III Rotation about an axis in rotation

Rotation about a fixed axis

Rotation of ceiling fan, potter's wheel, opening and closing of doors and needles of a wall clock etc. come into this category.

When a ceiling fan rotates, the vertical rod supporting it remains stationary and all the particles on the fan move on circular paths. Circular path of a particle P on one of its blades is shown by dotted circle. Centers of circular paths followed by every particle are on the central line through the rod. This central line is known as *axis of rotation* and is shown by a dashed line. All the particles on the axis of rotation are at rest, therefore the axis is stationary and the fan is in rotation about this fixed axis.

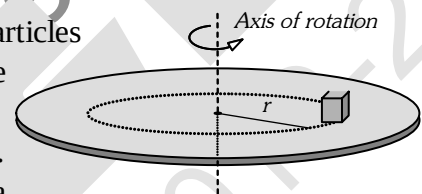


A door rotates about a vertical line that passes through its hinges. This vertical line is the axis of rotation. In the figure, the axis of rotation is shown by dashed line.

Axis of rotation

An imaginary line perpendicular to plane of circular paths of particles of a rigid body in rotation and containing the centers of all these circular paths is known as *axis of rotation*.

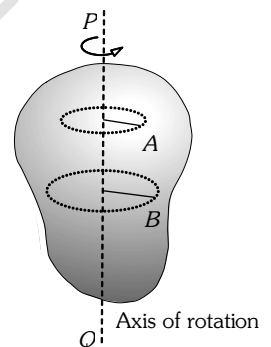
It is not necessary that the axis of rotation pass through the body. Consider system shown in the figure, where a block is fixed on a rotating disk. The axis of rotation passes through the center of the disk but not through the block.



Important observations

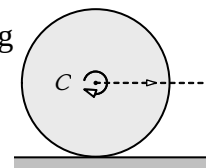
Let us consider a rigid body of arbitrary shape rotating about a fixed axis PQ passing through the body. Two of its particles A and B are shown moving on their circular paths.

- ⇒ All of its particles, not on the axis of rotation, move on circular paths with centers on the axis of rotation. All these circular paths are in parallel planes that are perpendicular to the axis of rotation.
- ⇒ All the particles of the body cover same angular displacement in the same time interval, therefore all of them move with the same angular velocity and angular acceleration.
- ⇒ Particles moving on circular paths of different radii move with different speeds and different magnitudes of linear acceleration. Furthermore, no two particles in the same plane perpendicular to the axis of rotation have same velocity and acceleration vectors.
- ⇒ All the particles on a line parallel to the axis of rotation move circular paths of the same radius therefore have same velocity and acceleration vectors.
- ⇒ Consider two particles in a plane perpendicular to the rotational axis. Every such particle on a rigid body in rotation motion moves on circular path relative to another one. Radius of the circular path equals to the distance between the particles. In addition, angular velocity and angular acceleration equals to that of rotation motion of the body.



Rotation about an axis in translation

Rotation about an axis in translation includes a broad category of motions. Rolling is an example of this kind of motion. A rod lying on table when pushed from its one in its perpendicular direction also executes this kind of motion. To understand more let us discuss few examples.



Consider rolling of wheels of a vehicle, moving on straight level road. Relative to a reference frame, moving with the vehicle wheel appears rotating about its stationary axel. The rotation of the wheel from this frame is rotation about fixed axis. Relative to a reference frame fixed with the ground, the wheel appears rotating about the moving axel, therefore, rolling of a wheel is superposition two simultaneous but distinct motions – rotation about the axel fixed with the vehicle and translation of the axel together with the vehicle.

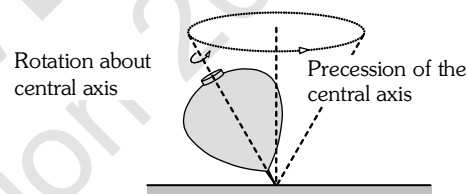
Important observations

- ⇒ Every particle of the body always remains in a plane perpendicular to the rotational axis. Therefore, this kind of motion is also known as *general plane motion*.
- ⇒ Relative to every particle another particle in a plane perpendicular to axis of rotation moves on circular path. Radius of the circular path equals to the distance between the particles and angular velocity and angular acceleration equals to that of rotation motion of the body.
- ⇒ Rotation about axis in translation is superposition of pure rotation about the axis and simultaneous translation motion of the axis.

Rotation about an axis in rotation.

In this kind of motion, the body rotates about an axis that also rotates about some other axis. Analysis of rotation about rotating axes is not in the scope of JEE, therefore we will discuss it to have an elementary idea only.

As an example consider a rotating top. The top rotates about its central axis of symmetry and this axis sweeps a cone about a vertical axis. The central axis continuously changes its orientation, therefore is in rotation motion. This type of rotation in which the axis of rotation also rotates and sweeps out a cone is known as *precession*.



Another example of rotation about axis in rotation is a table-fan swinging while rotating. Table-fan rotates about its horizontal shaft along which axis of rotation passes. When the rotating table-fan swings, its shaft rotates about a vertical axis.

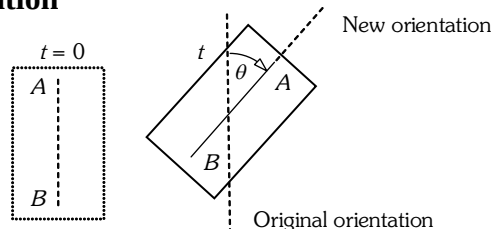
Angular displacement, angular velocity and angular acceleration

Rotation motion is the change in orientation of a rigid body with time. It is measured by turning of a linear dimension or a straight line drawn on the body.

In the figure is shown at two different instants $t = 0$ and t a rectangular plate moving in its own plane. Change in orientation during time t equals to the angle θ through which all the linear dimensions of the plate or a line AB turns.

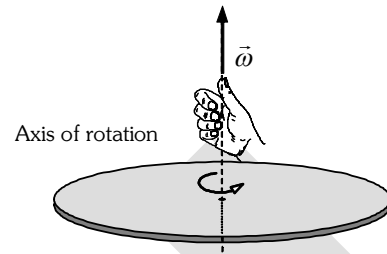
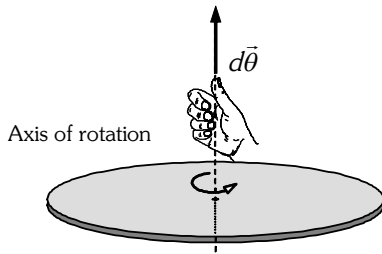
If the angle θ continuously changes with time t , instantaneous angular velocity ω and angular acceleration α for rotation of the body are defined by the following equations.

$$\omega = \frac{d\theta}{dt} \quad [1] \quad \alpha = \frac{d^2\theta}{dt^2} = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta} \quad [2]$$

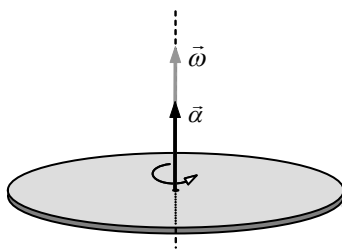
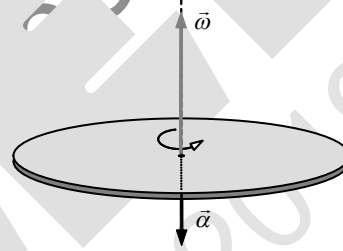


Direction of angular motion quantities

Angular displacement, angular velocity and angular acceleration are known as angular motion quantities. Infinitesimally small angular displacement, instantaneous angular velocity and angular acceleration are vector quantities. Direction of infinitesimally small angular displacement and instantaneous angular velocity is given by the right hand rule. For a disk rotating as shown in the figure, the angular velocity points upwards along the axis of rotation.



The direction of angular acceleration depends, whether angular velocity increases or decreases with time. For increasing angular velocity, the angular acceleration vector points in the direction of angular velocity vector and for decreasing angular velocity, the angular acceleration vector points opposite to the angular velocity vector.

**Angular acceleration: Increasing angular speed****Angular acceleration: Decreasing angular speed**

In rotation about fixed axis and rotation about axis in translation, the axis of rotation does not rotate and angular velocity and acceleration always point along the axis of rotation. Therefore, in dealing these kinds of motions, the angular motion quantities can be used in scalar notations by assigning them positive sign for one direction and negative sign for the opposite direction.

These quantities have similar mathematical relations as position coordinate, velocity, acceleration and time have in rectilinear motion.

⇒ A body rotating with constant angular velocity ω and hence zero angular acceleration is said to be uniform rotation. Angular position θ is given by equation

$$\theta = \theta_o + \omega t \quad [3]$$

⇒ Thus for a body rotating with uniform angular acceleration α , the angular position θ and angular velocity ω can be expressed by the following equation.

$$\omega = \omega_o + \alpha t \quad [4]$$

$$\theta = \theta_o + \omega_o t + \frac{1}{2} \alpha t^2 = \theta_o + \frac{1}{2} (\omega_o + \omega) t \quad [5]$$

$$\omega^2 = \omega_o^2 + 2\alpha(\theta - \theta_o) \quad [6]$$

Ex. A disk rotates about a fixed axis. Its angular velocity ω varies with time according to equation $\omega = at + b$. At the instant $t = 0$ its angular velocity is 1.0 rad/s at angular position is 2 rad and at the instant $t = 2$ s, angular velocity is 5.0 rad/s. Determine angular position θ and angular acceleration α when $t = 4$ s.

Sol. The given equation $\omega = at + b$ has form similar to eq.[4], therefore motion is rotation with uniform angular acceleration. Initial angular velocity $= \omega_o = b = 1.0 \text{ rad/s}$, Angular acceleration $\alpha = a$,

$$\theta = \frac{1}{2}at^2 + t + c$$

Since at $t = 0$, $\omega = 1.0$ rad/s, we obtain the constant c .

Initial angular position = $\theta_0 = c = 2.0 \text{ rad}$

Since at $t = 2.0$ s angular velocity is 5.0 rad/s, from given expression of angular velocity, we have

$$\omega = at + b \rightarrow$$

Substituting $b = 1.0 \text{ rad/s}$, $t = 2.0 \text{ s}$ and $\omega = 5.0 \text{ rad/s}$, we have $a = 2.0 \text{ rad/s}$

Now we can write expressions for angular position, angular velocity and angular acceleration.

$$\theta = t^2 + t + 2.0 \quad (1) \quad \omega = 2.0t + 1.0 \quad (2)$$

From the above equations, we can calculate angular position, angular velocity and angular acceleration at $t=4.0$ s

$$\theta_4 = 22 \text{ rad}, \omega_4 = 9.0 \text{ rad/s}, \alpha = 2.0 \text{ rad/s}^2 \quad \textbf{Ans.}$$

Kinematics of rotation about fixed axis

In figure is shown a rigid body of arbitrary shape rotating about the z -axis. In the selected frame (here the coordinate system) all the three axes are at rest, therefore the z -axis that is the axis of rotation is at rest and the body is in fixed axis rotation. All of its particles other than those on the z -axis move on circular paths with their centers on the z -axis. All these circular paths are parallel to the x - y plane. In the figure, one of its particles P is shown moving with velocity $\vec{v}$ on a circular path of radius r and center C . Its position vector is $\vec{R}$. It were at the line Cx at $t = 0$ and at the position shown at the instant t . During time interval t , it covers the circular arc of length s and its radius vector turns through angle θ .

In an infinitesimally small time interval dt let, the particle covers infinitesimally small distance ds along its circular path.

$$d\vec{s} = d\vec{\theta} \times \vec{r} = d\vec{\theta} \times \vec{R} \quad [7]$$

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{d\vec{r}}{dt} = \frac{d\vec{R}}{dt} \quad [8]$$

From eq. [7] and [8] we have

$$\vec{v} = \vec{\omega} \times \vec{r} = \vec{\omega} \times \vec{R} \quad [9]$$

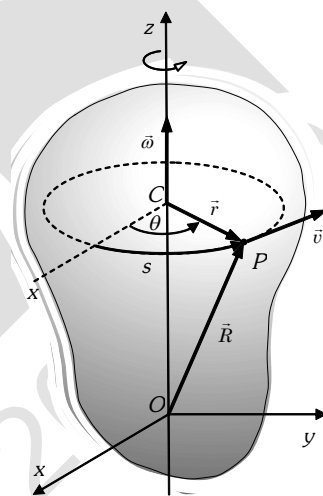
The above equation tells us the relation between the linear and angular velocity. Now we explore relation between the linear and angular accelerations. For the purpose, differentiate the above equation with respect to time.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} \quad [10]$$

The first term on the RHS points along the tangent in the direction of the velocity vector and it is known as tangential acceleration $\vec{a}_t$ same as we have in circular motion. In addition, the second term point towards the center C . It is known as centripetal acceleration of normal component $\vec{a}_n$ of acceleration same as in circular motion. Now we have

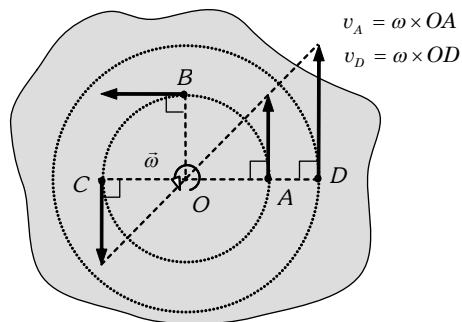
$$\text{Tangential acceleration} \quad \vec{a}_T = \vec{\alpha} \times \vec{r} \quad [11]$$

$$\text{Normal acceleration} \quad \vec{a}_n = \vec{\omega} \times \vec{v} = -\omega^2 \vec{r} \quad [12]$$



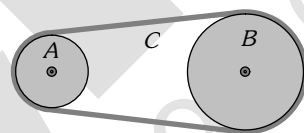
How to Locate Axis of Rotation

Every particle in a plane perpendicular to the axis of rotation move with different velocities and accelerations, moreover, they all have the same angular velocity and angular acceleration. Such a section of a body in rotation is shown here. The particles A , B and C at equal distance from the axis of rotation move with equal speeds v_A and the particle D moves with speed v_D on concentric circular paths. The location of rotational axis can be determined by any of the two graphical techniques.



- ⇒ Lines perpendicular to velocity vectors and passing through the particles, whose velocity vectors are neither parallel nor antiparallel intersect at the axis of rotation. See pairs of particles A and B , B and C and B and D .
- ⇒ Lines perpendicular to velocity vectors and passing through the particles, whose velocity vectors are either parallel or antiparallel, coincide and intersect the line joining tips of their velocity vectors at the axis of rotation. Refer pairs of particles A and C , A and D and C and D .

Ex. A belt moves over two pulleys A and B as shown in the figure. The pulleys are mounted on two fixed horizontal axels. Radii of the pulleys A and B are 50 cm and 80 cm respectively. Pulley A is driven at constant angular acceleration 0.8 rad/s^2 until the pulley B acquires an angular velocity of 10 rad/s . The belt does not slide on either of the pulleys.



- (a) Find acceleration of a point C on the belt and angular acceleration of the pulley B .
 (b) How long after the pulley B achieve angular velocity of 10 rad/s .

Sol. Since the belt does not slide on the pulleys, magnitude of velocity and acceleration of any point on the belt are same as velocity tangential acceleration of any point on periphery of either of the pulleys.

Using the above fact with eq.[11], we have

$$\vec{a}_T = \vec{\alpha} \times \vec{r} \rightarrow$$

$$a_C = \alpha_A r_A = \alpha_B r_B$$

Substituting $r_A = 0.5 \text{ m}$, $r_B = 0.8 \text{ m}$ and $\alpha = 0.8 \text{ rad/s}^2$, we have

$$a_C = 5 \text{ m/s}^2 \text{ and } \alpha_B = \frac{a_C}{r_B} = \frac{\alpha_A r_A}{r_B} = 0.5 \text{ rad/s}^2 \quad \text{Ans.}$$

From eq. [4], we have

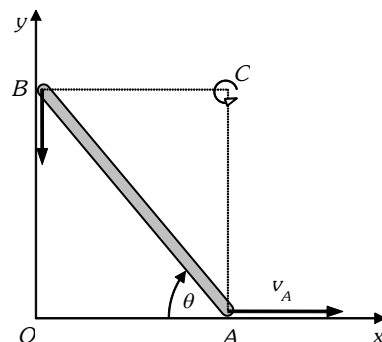
$$\omega = \omega_o + \alpha t \rightarrow t = \frac{\omega_B - \omega_{Bo}}{\alpha_B}$$

Substituting $\omega_{Bo} = 0$, $\omega_B = 10 \text{ rad/s}$ and $\alpha_B = 0.5 \text{ rad/s}^2$, we have $t = 20 \text{ s}$ **Ans.**

Kinematics of rotation about axis in translation

In this kind of motion, the body rotates about an axis and the axis moves without rotation. Rolling is a very common example of this kind of motion.

As an example consider a rod whose ends A and B are sliding on the x and y -axis as shown in the figure. Change in its orientation measured by change in angle θ indicates that the rod is in rotation. Perpendiculars drawn to velocity vector of its end points intersect at the axis of rotation, which is continuously changing its position.

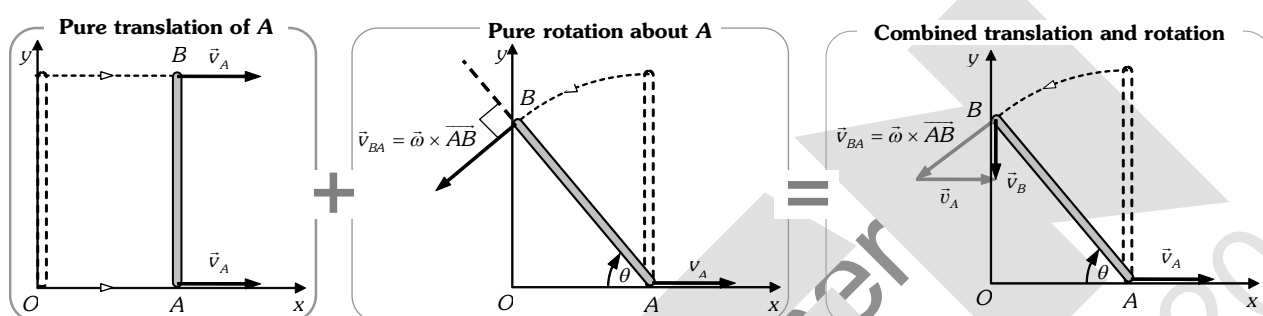


Instantaneous Axis of Rotation (IAR)

It is a mathematical line about that a body in combined translation and rotation can be conceived in pure rotation at an instant. It continuously changes its location.

Now we explore how the combined translation and rotational motion of the rod is superposition of translation motion of any of its particle and pure rotation about an axis through that particle.

Consider motion of the rod from beginning when it was parallel to the y -axis. In the following figure translation motion of point A is superimposed with pure rotation about A .



The motion of the rod can be conceived as superposition of translation of point A and simultaneous rotation about an axis through A.

The same experiment can be repeated to demonstrate that motion of the rod can be conceived as superposition of translation of any of its particle and simultaneous rotation about an axis through that particle.

Considering translation of A and rotation about A this fact can be expressed by the following equation.

Combined Motion = Translation of point A + Pure rotation about point A

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \quad [13]$$

Since point B moves relative A moving on circular path its velocity relative to A is given by the equation

$$\vec{v}_{BA} = \vec{\omega} \times \overrightarrow{AB}.$$

Now we have

$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \overrightarrow{AB} \quad [14]$$

The above fact is true for any rigid body in combined translation and rotation motion.

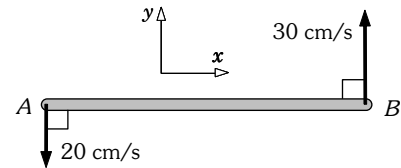
Rotation about an axis in translation of a rigid body can be conceived as well as analyzed as superposition of translation motion of any of its particle and simultaneous rotation about an axis passing through that particle provided that the axis is parallel to the actual one.

Similar to eq.[13], we can write equation for acceleration.

$$\left. \begin{aligned} \vec{a}_A &= \vec{a}_B + \vec{a}_{BA} \\ \vec{a}_A &= \vec{a}_B + \overbrace{\vec{a}_{BAT} + \vec{a}_{BA_n}} \\ \vec{a}_A &= \vec{a}_B + \vec{\alpha} \times \overline{AB} + -\omega^2 \overline{AB} \end{aligned} \right\} \quad [15]$$

Ex. A 100 cm rod is moving on a horizontal surface. At an instant, when it is parallel to the x-axis its ends A and B have velocities 30 cm/s and 20 cm/s as shown in the figure.

- (a) Find its angular velocity and velocity of its center.
(b) Locate its instantaneous axis of rotation.



Sol.

Let the rod is rotating anticlockwise, therefore its angular velocity is given by $\vec{\omega} = \omega \hat{k}$. Velocity vectors of all the points on the rod and its angular velocity must satisfy the relative motion eq.[14].

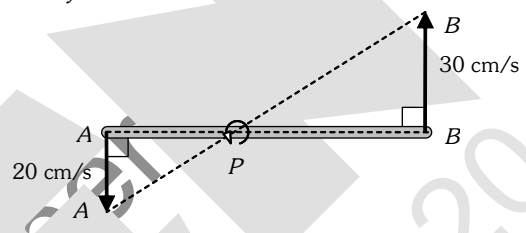
- (a) Substituting velocities $\vec{v}_A = -20\hat{j}$ cm/s and $\vec{v}_B = 30\hat{j}$ cm/s and angular velocity $\vec{\omega}$ in eq.[14], we

$$\text{have } \vec{v}_B = \vec{v}_A + \vec{\omega} \times \overline{AB} \rightarrow \omega = 0.5 \text{ rad/s} \quad \text{Ans.}$$

Velocity vector of the center C of the rod also satisfy the following equation.

$$\vec{v}_C = \vec{v}_A + \vec{\omega} \times \overline{AC} \rightarrow \vec{v}_C = -20\hat{j} + 0.5\hat{k} \times 50\hat{i} = 5.0\hat{j} \text{ cm/s} \quad \text{Ans.}$$

- (b) Here velocity vectors of the particles A and B are antiparallel, therefore the instantaneous axis of rotation passes through intersection of the common perpendicular to their velocity vectors and a line joining tips of the velocity vectors. The required geometrical construction is shown in the following figure.



Since triangles AA'P and BB'P are similar and $AB = 100$ cm, we have $AP = 40$ cm.

The instantaneous axis of rotation passes through the point P, which is 40 cm from A. **Ans.**

Analytical Approach.

The instantaneous center of rotation is at instantaneous rest. Using this fact in eq.[14], we have

$$\vec{v}_P = \vec{v}_A + \vec{\omega} \times \overline{AP} \rightarrow \vec{0} = -20\hat{j} + 0.5\hat{k} \times (AP)\hat{j} \Rightarrow AP = 40 \text{ cm} \quad \text{Ans.}$$

Concept of Rotational Inertia (Moment of inertia)

Total mass of a body in translation motion is the measure of its inertia to translation motion. Similarly if a point mass m is rotating about an axis at a distance r from the axis, then term mr^2 provides suitable measure of its inertia to rotation motion. *The inertia to rotation motion is known as rotational inertia or more commonly moment of inertia.*

Moment of inertia of a rigid body

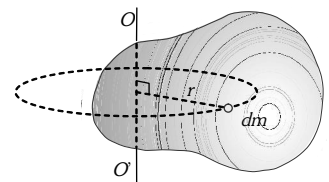
A rigid body is continuous distribution of mass and can be assumed consisting of infinitely large number of point particles. If one of the point particle of infinitely small mass dm is at a distance r from the axis of rotation OO' , the moment of inertia of this point particle is given by

$$dI_o = r^2 dm$$

The moment of inertia of the whole body about the axis OO' can now be obtained by integrating term of the above equation over the limits to cover whole of the body.

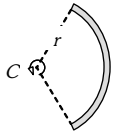
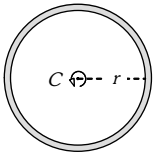
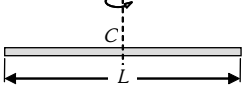
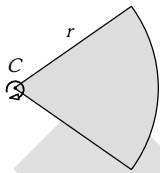
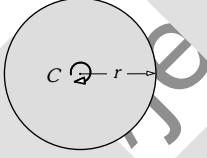
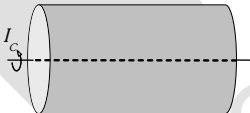
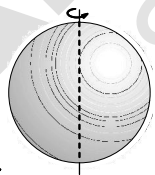
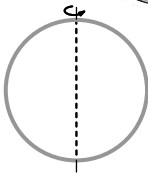
$$I_o = \int dI_o = \int r^2 dm$$

Expression for moment of inertia contains product of two terms. One of them is the mass of the body and the other is a characteristic dimension, which depends on the manner how mass of the body is distributed relative to the axis of rotation. Therefore moment of inertia of a rigid body depends on the mass of the body and distribution of the mass relative to the axis of rotation. Obviously for uniform bodies expression of moment of inertia depends on their shape and location and orientation of the axis of rotation. Based on these facts we can conclude



1. If mass distribution is similar for two bodies about an axis, expressions of their moment of inertia must be of the same form about that axis.
2. If the whole body or any of its portions is shifted parallel to the axis of rotation, moment of inertia remains unchanged.

Moment of Inertia for some commonly used bodies

Body	Axis	Moment of Inertia
Uniform thin rod bent into shape of an arc of mass m		Passing through center and perpendicular to the plane containing the arc $I_C = mr^2$
Uniform ring of mass m		Passing through center and perpendicular to the plane containing the arc or the centroidal axis. $I_C = mr^2$
Straight uniform rod		Passing through center and perpendicular to the rod or the centroidal axis. $I_C = \frac{mL^2}{12}$
Sector of a uniform disk of mass m		Passing through center and perpendicular to the plane containing the sector. $I_C = \frac{mr^2}{2}$
Uniform disk of mass m		Passing through center and perpendicular to the plane containing the disk or the centroidal axis. $I_C = \frac{mr^2}{2}$
Homogeneous cylinder of mass m		Axis of the cylinder or the centroidal axis. $I_C = \frac{mr^2}{2}$
Homogeneous sphere of mass m		Diameter or the centroidal axis $I_C = \frac{2}{5}mR^2$
Spherical shell of mass m		Diameter or the centroidal axis $I = \frac{2}{3}mR^2$

Theorems on Moment of Inertia

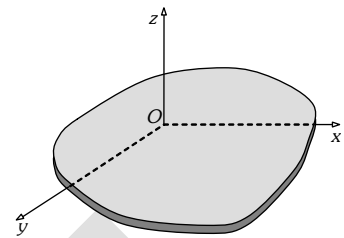
Moment of inertias of a rigid body about different axes may be different. There are two theorems known as *theorem of perpendicular axes* and *theorem of parallel axes*, which greatly simplify calculation of moment of inertia about an axis if moment of inertia of a body about another suitable axis is known.

Theorem of Perpendicular Axes

This theorem is applicable for a rigid body that lies entirely within a plane i.e. a laminar body or a rod bent into shape of a plane curve. The moment of inertia I_x , I_y and I_z of the body about the x , y and z -axis can be expressed by the following equations.

$$I_z = I_y + I_x$$

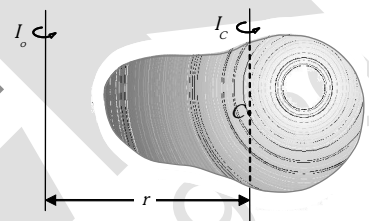
For a planar body, the moment of inertia about an axis perpendicular to the plane of the body is the sum of the moment of inertias about two perpendicular axes in the plane of the object provided that all the three axes are concurrent.

**Theorem of Parallel Axes**

This theorem also known as Steiner's theorem can be used to determine the moment of inertia of a rigid body about any axis, if the moment of inertia of the body about a parallel axis passing through mass center of the body and perpendicular distance between both the axes is known.

Consider a body of arbitrary shape and mass m shown in the figure. Its moment of inertia I_o and I_c are defined about two parallel axes. The axis about which moment of inertia I_c is defined passes through the mass center C . Separation between the axes is r . These two moment of inertias are related by the following equation.

$$I_o = I_c + Mx_c^2$$



The above equation is known as the theorem of parallel axes or Steiner's theorem.

- ⇒ The moment of inertia about any axis parallel to an axis through the mass center is given by sum of moment of inertia about the axis through the mass center and product term of mass of the body and square of the distance between the axes.
- ⇒ Among all the parallel axes the moment of inertia of a rigid body about the axis through the mass center is the minimum moment of inertia.

The second term added to the moment of inertia I_c about the centroidal axis in the above equation can be recognized as the moment of inertia of a particle of mass equal to that of the body and located at its mass center. It again reveals that the plane motion of a rigid body is superposition of pure rotation about the mass center or centroidal rotation and translation of its mass center.

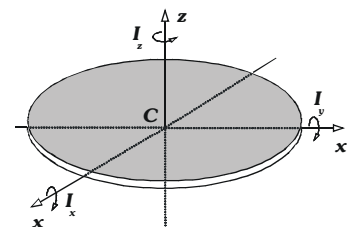
Ex. Find moment of inertia of a uniform disk of mass m and radius r about one of its diameter.

Sol. In the adjoining figure a disk is shown with two of its diameter perpendicular to each other. These diameters are along the x and the y -axis of a coordinate system. The x -axis is perpendicular to the plane of the disk and passes through its center is also shown. Since the disk is symmetric about both the diameters, moment of inertias about both the diameters must be equal. Thus substituting this in the theorem of perpendicular axes, we have

$$I_z = I_y + I_x \rightarrow I_z = 2I_x = 2I_y$$

Moment of inertia of the disk about the z -axis is $I_z = \frac{1}{2}mr^2$. Substituting it in the above equation, we have

$$I_x = I_y = \frac{1}{2}I_z = \frac{1}{4}mr^2 \quad \text{Ans.}$$



Radius of Gyration

It is the radial distance from a rotation axis at which the mass of an object could be concentrated without altering the moment of inertia of the body about that axis.

If the mass m of the body were actually concentrated at a distance k from the axis, the moment of inertia about that axis would be mk^2 .

$$k = \sqrt{\frac{I}{m}}$$

The radius of gyration has dimensions of length and is measured in appropriate units of length such as meters.

DYNAMICS OF RIGID BODY

Torque: Moment of a force

Torque is rotational analogue of force and expresses tendency of a force applied to an object to cause the object to rotate about a given point.

To investigate further let us discuss an experiment. Consider a rod pivoted at the point O . A force $\vec{F}$ is applied on it at the point P . The component $F \cos \theta$ of the force along the rod is counterbalanced by the reaction force of the pivot and cannot contribute in rotating the rod. It is the component $F \sin \theta$ of the force perpendicular to the rod, which is responsible for rotation

of the rod. Moreover, farther is the point P from O , where the force is applied easier is to rotate the rod. This is why handle on a door is attached as far away as possible from the hinges.

Magnitude of torque of a force is proportional to the product of distance of point of application of the force from the pivot and magnitude of the perpendicular component $F \sin \theta$ of the force. Denoting torque by symbol τ , the distance of point of application of force from the pivot by r , we can write

$$\tau_o \propto rF \sin \theta$$

Since rotation has sense of direction, torque should also be a vector. Its direction is given by right hand rule. Now we can express torque by the cross product of $\vec{r}$ and $\vec{F}$.

$$\vec{\tau}_o = \vec{r} \times \vec{F}$$

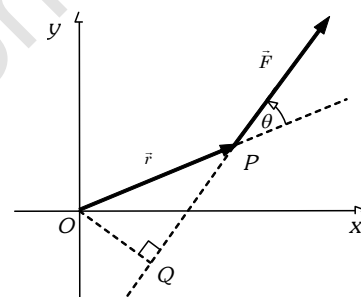
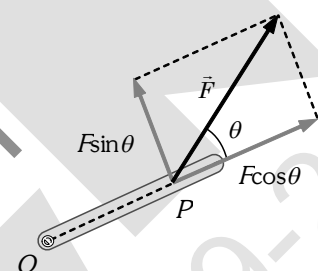
Here constant of proportionality has been assumed a dimensionless number unity because a unit of torque has been chosen as product of unit of force and unit of length.

The geometrical construction shown in figure suggests a simple way to calculate torque. The line OQ ($= r \sin \theta$) known as *moment arm*, is the length of perpendicular drawn from O on the line of action of the force. The magnitude of the torque equals to the product of OQ and magnitude of the force $\vec{F}$.

Torque about a Point and Torque about an Axes

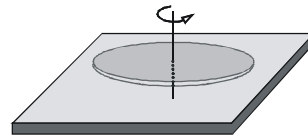
We have defined *torque of a force about a point* as the moment of the force about that point. In dealing with rotation about a fixed axis we need to know torque about the axis rotation.

When a body is in plane motion the net torque of all the forces including the forces necessary to restrain rotation of the axis is along the axis of rotation. It is known as torque about the axis. Torque of a force about an axis of rotation equals to the moment of force about the point where plane of motion of the point of application of the force intersects the axis.

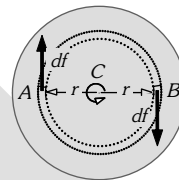


In analyzing plane motion we always consider torque about an axis under consideration and in rest of the book by the term torque of force we mean torque about an axis.

Ex. A uniform disk of mass M and radius R rotating about a vertical axis passing through its center and perpendicular to its plane is placed gently on a rough horizontal ground, where coefficient of friction is μ . Calculate torque of the frictional forces.



Sol. When the disk rotates on the ground, kinetic friction acts at every contact point. Since the gravity acts uniformly everywhere and the disk is also uniform, the normal reaction from the ground is uniformly distributed over the entire contact area. Consider two diametrically opposite identical portions A and B of the disk each of mass dm at distance r from the center as shown in the adjacent figure. The normal reaction from the ground on each of these portions equals to their weights and hence frictional forces are $df = \mu dm g$.



Consider a ring of radius r and width dr shown by dashed lines. Net torque $d\tau_c$ of friction force on this ring can easily be expressed by the following equation.

$$d\tau_c = r\mu(\text{mass of the ring})g = r\mu\left(\frac{\text{Mass of the disk}}{\text{Area of the disk}} \times \text{Area of the ring}\right)g = r\mu\left(\frac{2Mrdr}{R^2}\right)g$$

Integrating both sides of the above equation, we have

$$\tau_c = \frac{2\mu Mg}{R^2} \int_{r=0}^R r^2 dr = \frac{2}{3} \mu MgR \quad \text{Ans.}$$

Rotational equilibrium

A rigid body is said to be state of rotational equilibrium if its angular acceleration is zero. Therefore a body in rotational equilibrium must either be in rest or rotation with constant angular velocity.

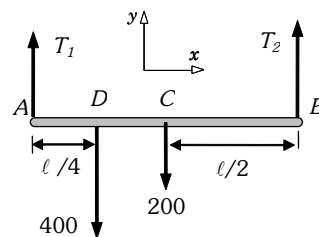
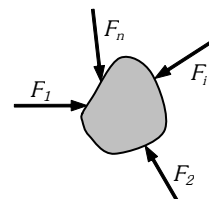
Since scope of JEE syllabus is confined only to rotation about a fixed axis or rotation about an axis in translation motion, the discussion regarding rotational equilibrium is limited here to situations involving only coplanar forces. Under these circumstances the necessary and sufficient condition for rotational equilibrium is

If a rigid body is in rotational equilibrium under the action of several coplanar forces, the resultant torque of all the forces about any axis perpendicular to the plane containing the forces must be zero.

In the figure a body is shown under the action of several external coplanar forces $F_1, F_2, \dots, F_i$, and F_n .

$$\sum \tau_P = 0$$

Here P is a point in the plane of the forces about which we calculate torque of all the external forces acting on the body. The flexibility available in selection of the point P provides us with advantages that we can select such a point about which torques of several unknown forces will become zero or we can make as many number of equations as desired by selecting several different points. The first situation yields to a simpler equation to be solved and second situation though does not give independent equation,



which can be used to determine additional unknowns yet may be used to check the solution.

The above condition reveals that a body cannot be in rotational equilibrium under the action of a single force unless the line of action passes through the mass center of the body.

A case of particular interest arises where only three coplanar forces are involved and the body is in rotational equilibrium. It can be shown that *if a body is in rotational equilibrium under the action of three forces, the lines of action of the three forces must be either concurrent or parallel*. This condition provides us with a graphical technique to analyze rotational equilibrium.

Equilibrium of Rigid Bodies

A rigid body is said to be in equilibrium, if it is in translational as well as rotational equilibrium both. To analyze such problems conditions for both the equilibriums must be applied.

Ex. A uniform rod of 20 kg is hanging in horizontal position with the help of two threads. It also supports a 40 kg mass as shown in the figure. Find the tension developed in each thread.

Sol. Free body diagram of the rod is shown in the figure.

Translational equilibrium

$$\Sigma F_y = 0 \rightarrow T_1 + T_2 = 400 + 200 = 600 \text{ N}$$

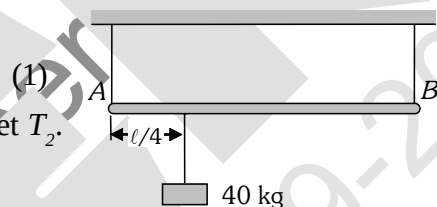
Rotational equilibrium: Applying the condition about A, we get T_2 .

$$\Sigma \vec{\tau}_A = \vec{0} \rightarrow 400(l/4) + 200(l/2) - T_2 l = 0$$

$$T_2 = 200 \text{ N} \quad \text{Ans.}$$

Similarly writing torque equation about B, we have

$$\Sigma \vec{\tau}_B = \vec{0} \rightarrow T_1 = 400 \text{ N. Ans.}$$



Force and Torque equations in General Plane Motion

Motion of a rigid body either pure rotation or rotation about axis in translation can be thought and analyzed as superposition of translation of any of its particle and simultaneous rotation about an axis passing through that particle provided that the axis remain parallel to the original one. As far as kinematics is concerned this particle may or may not be the mass center. Whereas in dealing with kinetics, general plane motion is conceived as superposition of translation motion of the mass center and simultaneous centroidal rotation.

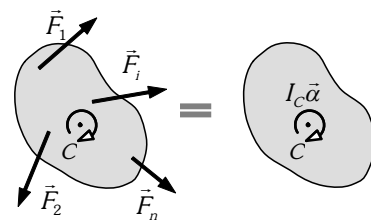
To make use the above idea and equations developed in the previous section we classify pure rotation i.e. rotation about fixed axis into two categories and deal with general plane motion as the third category.

Pure centroidal rotation: Rotation about fixed axis through mass centre

In this kind of rotation motion the axis of rotation passes through the mass center and remain fixed in space. Rotation of ceiling fan is a common example of this category. It is a subcategory of pure rotation. The axis of rotation passes through the mass center and remains fixed. In this kind of rotation the mass center of the body does not move.

In the figure, free body diagram and kinetic diagram of a body rotating about a fixed axis passing through its mass center C is shown. The mass center of the body does not accelerate; therefore we only need to write the torque equation.

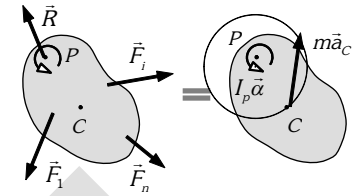
$$\Sigma \vec{\tau}_C = I_C \vec{\alpha}$$



Rotation about fixed axis not passing through mass center

In this kind of rotation the axis of rotation remains fixed and does not pass through the mass center. Rotation of door is a common example of this category. Doors are hinged about their edges; therefore their axis of rotation does not pass through the mass center. In this kind of rotation motion the mass center executes circular motion about the axis of rotation.

In the figure, free body diagram and kinetic diagram of a body rotating about a fixed axis through point P is shown. It is easy to conceive that as the body rotates its mass center moves on a circular path of radius $\vec{r}_{P/C}$. The mass center of the body is in translation motion with acceleration $\vec{a}_C$ on circular path of radius $r_{P/C}$. To deal with this kind of motion, we have to make use of both the force and the torque equations.



Translation of mass center

$$\Sigma \vec{F}_i = M\vec{a}_C = M\vec{\alpha} \times \vec{r}_{C/P} - M\omega^2 \vec{r}_{C/P}$$

Centroidal Rotation

$$\Sigma \vec{\tau}_C = I_C \vec{\alpha}$$

Making use of parallel axis theorem

$$(I_P = Mr_{P/C}^2 + I_C) \text{ and } \vec{a}_{C/P} = \vec{\alpha} \times \vec{r}_{C/P} - \omega^2 \vec{r}_{C/P}$$

we can write the following equation also.

Pure Rotation about P

$$\Sigma \vec{\tau}_P = I_P \vec{\alpha}$$

General Plane Motion: Rotation about axis in translation motion

Rotation of bodies about an axis in translation motion can be dealt with either as superposition of translation of mass center and centroidal rotation or assuming pure rotation about the instantaneous axis of rotation. In the figure is shown the free body diagram and kinetic diagram of a body in general plane motion.

Translation of mass center

$$\sum_{i=1}^n \vec{F}_i = M\vec{a}_C$$

Centroidal Rotation

$$\sum_{i=1}^n \vec{\tau}_C = I_C \vec{\alpha}$$

This kind of situation can also be dealt with considering it rotation about IAR. It gives sometimes quick solutions, especially when IAR is known and forces if acting at the IAR are not required to be found.

Example 10

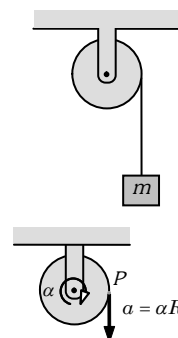
A block of mass m is suspended with the help of a light cord wrapped over a cylindrical pulley of mass M and radius R as shown in the figure. The system is released from rest. Find the angular acceleration of the pulley and the acceleration of the block.

Solution.

After the system is released, the block is in translation motion and the pulley in rotation about an axis passing through its mass center i.e. in pure rotation.

Let the block moves vertically down with acceleration a pulling the cord down and causing the pulley to rotate clockwise. Since the cord is inextensible every point on its vertical portion and point of contact P of the pulley move down with acceleration a as shown in the adjacent figure. It is the tangential acceleration of point P so the angular acceleration α of the pulley rotating in clockwise sense is given by

$$a = \alpha R \quad (1)$$



The forces acting on the pulley and on the block are shown in their free-body diagrams along with the effective torque $I_C \alpha$ of the pulley and effective force ma of the block. Here T is the tension in the string, R is the reaction by the axel of the pulley, Mg is weight of the pulley and mg is weight of the block.

The pulley is in rotation about fixed axis through its mass center so we use eq. .

$$\sum \vec{\tau}_C = I_C \vec{\alpha} \rightarrow TR = I_C \alpha$$

After substituting $I_C = \frac{1}{2}MR^2$ and α from eq. (1), we have

$$T = \frac{1}{2}Ma \quad (2)$$

The block is in translation motion, so we use Newton's second law

$$\sum \vec{F} = m\vec{a} \rightarrow mg - T = ma \quad (3)$$

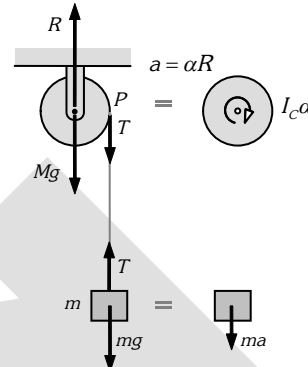
From equation (2) and (3), we have

$$\text{Acceleration of the block} \quad a = \frac{2mg}{M+2m}$$

Ans.

From eq. (1) and the above, we have

$$\alpha = \frac{2mg}{R(M+2m)} \quad \text{Ans.}$$



Ex. A thread is wrapped around a uniform disk of radius r and mass m . One end of the thread is attached to a fixed support on the ceiling and the disk is held stationary in vertical plane below the fixed support as shown in the figure. When the disk is set free, it rolls down due to gravity. Find the acceleration of the center of the disk and tension in the thread.

Sol. The point P , where the thread leaves the disk is always at instantaneous rest; therefore the disk can be assumed rolling without slipping with ICR at point P . Acceleration of the mass center $\vec{a}_C$ and angular acceleration of the disk are shown in the adjacent figure. Applying condition for rolling on stationary surface, we have

$$\vec{a}_C = \vec{\alpha} \times \vec{r}_{C/P} \rightarrow a_C = \alpha r \quad (1)$$

The disk rolls down on the vertical stationary thread. Its motion can either be analyzed as superposition of translation of the mass center and simultaneous centroidal rotation or a pure rotation about ICR. Since tension, which acts at the ICR is asked; we prefer superposition of translation of the mass center and simultaneous centroidal rotation.

Forces acting on the disk are tension T applied by the thread at point P and weight of the disk. These forces and the effective force ma_C and effective torque $I_C \alpha$ are shown in the adjacent figure.

Applying Newton's second law for translation of mass center, we have

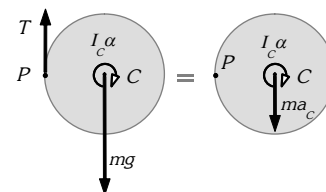
$$\sum \vec{F}_i = M\vec{a}_C \rightarrow mg - T = ma_C \quad (2)$$

Applying torque equation for centroidal rotation, we have

$$\sum \vec{\tau}_C = I_C \vec{\alpha} \rightarrow Tr = I_C \alpha$$

Substituting $\frac{1}{2}mr^2$ for I_C and α from eq. (1), we have

$$T = \frac{1}{2}ma_C \quad (3)$$



From eq. (2) and (3), we have

Acceleration of the mass center $a_C = \frac{2}{3}g$ **Ans.**

Tension in the string $T = \frac{1}{3}mg$ **Ans.**

Energy Methods

Newton's laws of motion tell us what is happening at an instant, while method of work and energy equips us to analyze what happens when a body moves from one place to other or a system changes its configuration. In this section, we introduce how to use methods of work and energy to analyze motion of rigid bodies.

Concept of Work in rotation motion

Work of a force is defined as the scalar product of the force vector and displacement vector of the point of application of the force. If during the action of a force $\vec{F}$ its point of application moves from position $\vec{r}_1$ to $\vec{r}_2$, the work $W_{1 \rightarrow 2}$ done by the force is expressed by the following equation.

$$W_{1 \rightarrow 2} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

Either we can use of this idea to calculate work of a force or its modified

version in terms of torque and angular displacement.

The work done by a torque during a finite rotation of the rigid body from initial value θ_i of the angle θ to final value θ_f , can be obtained by integrating both the sides of the equation given

$$W_{i \rightarrow f} = \int_{\theta_i}^{\theta_f} \vec{\tau}_O \cdot d\vec{\theta}$$

Kinetic Energy of a rigid body in rotation motion

A rigid body can be represented as a system of large number of particles, which keep their mutual distances unchanged in all circumstances. Kinetic energy of the whole body must be sum of kinetic energies of all of its particles. In this section we develop expressions for kinetic energy of a rigid body.

Kinetic Energy of a rigid body in plane motion

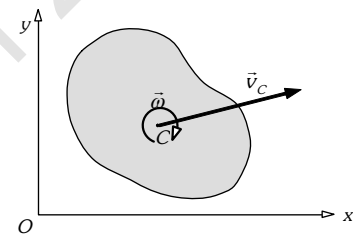
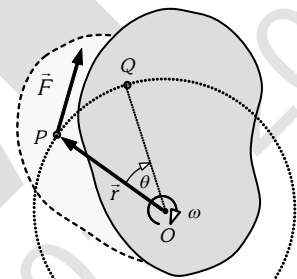
In the figure is shown a body in plane motion. Its mass center at an instant is moving with velocity $\vec{v}_C$ and rotating with angular velocity $\vec{\omega}$. Both these motions are shown superimposed in the given figure.

Kinetic energy too can be written as sum of kinetic energy ($\frac{1}{2}Mv_C^2$) due to translation motion of the mass center and kinetic energy ($\frac{1}{2}I_C\omega^2$) due to centroidal rotation.

$$K = \frac{1}{2}Mv_C^2 + \frac{1}{2}I_C\omega^2$$

If location of the instantaneous axis of rotation (IAR) is known, making use of the parallel axis theorem we can write kinetic energy by the following equation also.

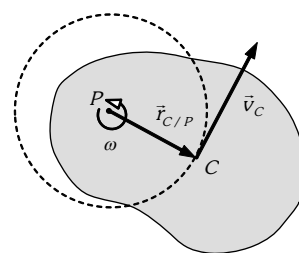
$$K = \frac{1}{2}I_{IAR}\omega^2$$



Kinetic Energy of a rigid body in rotation about fixed axis not passing through the mass centre

In this kind of motion the mass center is in circular motion about the axis of rotation. In the figure is shown a body rotation with angular velocity ω about a fixed axis through pint P and perpendicular to plane of the paper.

Mass center moves with speed $v_c = \omega r$. Kinetic energy of the body can now be expressed by the following equation.



$$K = \frac{1}{2} M v_C^2 + \frac{1}{2} I_C \omega^2$$

Making use of the parallel axis theorem ($I_P = I_{P/C} + I_C$) we can write kinetic energy by the following equation also.

$$K = \frac{1}{2} I_P \omega^2$$

Kinetic Energy of a rigid body in pure centroidal rotation

In pure centroidal rotation the mass center remain at rest; therefore kinetic energy due to translation of mass center vanishes.

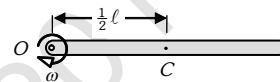
$$K = \frac{1}{2} I_C \omega^2$$

Ex. A rod of mass m and length ℓ is pivoted to a fixed support at one of its ends O . It is rotating with constant angular velocity ω . Write expression for its kinetic energy.



Sol. If the point C is the mass center of the rod, from theorem of parallel axes, the moment of inertia I_o of the rod about the fixed axis is

$$I_O = I_C + m(OC)^2 \rightarrow I_O = I_C + \frac{1}{4}m\ell^2$$



Substituting $\frac{1}{12}m\ell^2$ for I_c , we have

$$I_O = \frac{1}{3} m \ell^2$$

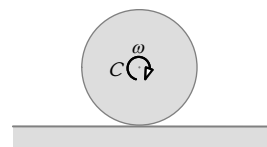
Kinetic energy of the rod equals to kinetic energy due to rotation about the fixed axis.

$$K = \frac{1}{2} I_{\omega} \omega^2 \rightarrow$$

Using above expression for I_ϕ , we have

$$K = \frac{1}{2} m \ell^2 \omega^2 \quad \text{Ans.}$$

Ex. A uniform rigid body of mass m and round section of radius r is rolling on horizontal ground with angular velocity ω . Its radius of gyration about the centroidal axis is k .



(a) Write expression of its kinetic energy.

(b) Also express the kinetic energy as sum of kinetic energy due to translation of mass center and kinetic energy due to simultaneous centroidal rotation.

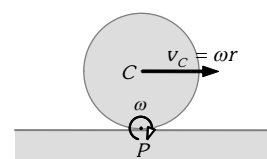
Sol. (a) The point of contact with ground of a body rolling on the ground is its ICR. Let the point P is the ICR as shown in the adjacent figure. The geometrical center C of a uniform body and the mass center coincide. Therefore moment of inertia I_P of the body about the ICR can be written by using the theorem of parallel axes.

$$I_P = I_C + m(PC)^2 \rightarrow$$

$$I_P = I_C + mr^2$$

Substituting $I_C = mk^2$, we have

$$I_p = m(k^2 + r^2) \quad (1)$$



Kinetic energy of a rigid body equals to kinetic energy due to rotation about the ICR.

$$K = \frac{1}{2} I_P \omega^2 \rightarrow \text{Substituting } I_P \text{ from eq. (1), we have}$$

$$K = \frac{1}{2} m(k^2 + r^2) \omega^2 \quad \text{Ans.}$$

- (b) Kinetic energy of the body also equals to sum of kinetic energy due to translation of its mass center and kinetic energy due to simultaneous centroidal rotation.

$$K = \frac{1}{2} m v_C^2 + \frac{1}{2} I_C \omega^2 \rightarrow \text{Substituting condition for rolling } v_C = \omega r \text{ and } I_C = m k^2, \text{ we have}$$

$$K = \frac{1}{2} m(\omega r)^2 + \frac{1}{2} m k^2 \omega^2 = \frac{1}{2} m(r^2 + k^2) \omega^2 \quad \text{Ans.}$$

Rolling as rotation about an axis in translation

If the point of contact of the of the rolling body does not slide it is known as *rolling without slipping* or *pure rolling* or simply *rolling* and if the point of contact slides it is known as *rolling with slipping*. All kind of rolling motion is examples of rotation about an axis in translation.

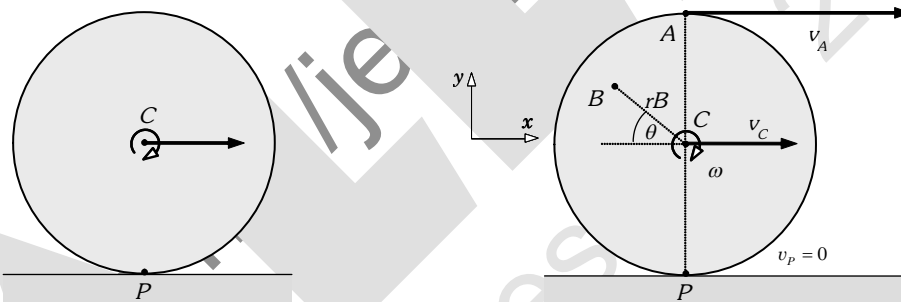
Rolling without slipping on stationary surface.

We first discuss velocity relations and thereafter accelerations relations of two points of a body of round section rolling on a stationary surface. For the purpose, we can use any of the following methods.

- | | | |
|----|-----------------------|---|
| I | Analytical Method: | By using relative motion equations. |
| II | Superposition Method: | By superimposing translation of a point and pure rotation about that point. |

Velocity relations by Analytical Method

Its point of contact P does not slide on the surface, therefore velocity of the point of contact relative to the surface is zero. In the next figure, velocity vectors of its center C and top point A are shown.



Velocity of the center C can be obtained with the help of relative motion equation.

$$\vec{v}_C = \vec{v}_P + \vec{\omega} \times \vec{PC} \rightarrow \vec{v}_C = \vec{0} + (-\omega \hat{k}) \times R \hat{j}$$

$$\vec{v}_C = \omega R \hat{i}$$

[16]

The above equation is used as condition of rolling without slipping on stationary surface.

Velocity of the top point A can be obtained by relative motion equation.

$$\vec{v}_A = \vec{v}_P + \vec{\omega} \times \vec{PA} \rightarrow \vec{v}_A = \vec{0} + (-\omega \hat{k}) \times (2R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

[17]

Once velocity of the center is obtained, we can use relative motion between A and C as well.

$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{CA} \rightarrow \vec{v}_A = \omega R \hat{i} + (-\omega \hat{k}) \times (R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

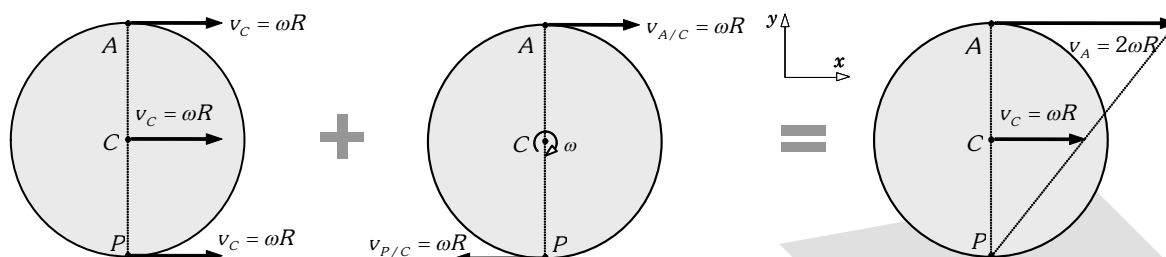
[18]

In similar fashion, velocity vector of an arbitrarily chosen point B .

$$\begin{aligned}\vec{v}_B &= \vec{v}_C + \vec{\omega} \times \overline{CB} \rightarrow \vec{v}_B = v_C \hat{i} + (-\omega \hat{k}) \times (-r \cos \theta \hat{i} + r \sin \theta \hat{j}) \\ \vec{v}_B &= (v_C + \omega r \sin \theta) \hat{i} + \omega r \cos \theta \hat{j}\end{aligned}\quad [19]$$

Velocity relations by Superposition Method

Now we will see that the above velocity relation can also be obtained by assuming rolling of the wheel as superposition of translation of its center and simultaneous rotation about the center.

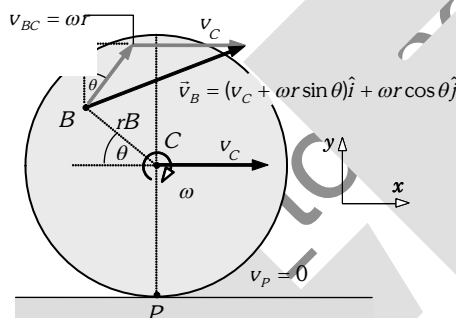


Translation of the center

Pure rotation about the center

Rolling

Velocity of an arbitrary point B as superposition of translation the center and rotation about the center.



Ex. A cylinder of radius 5 m rolls on a horizontal surface. Velocity of its center is 25 m/s. Find its angular velocity and velocity of the point A.

Sol. In rolling the angular velocity $\vec{\omega}$ and velocity of the center of a round section body satisfy condition described in the relative motion eq.[14].

So we have

$$\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P} \rightarrow 25\hat{j} = \omega\hat{k} \times 5\hat{j} \Rightarrow \vec{\omega} = -5\hat{k} \text{ rad/s}$$

Angular velocity vector points in the negative z-axis so the cylinder rotates in clockwise sense.

Velocity of the point A can be calculated by either analytical method, superposition method.

Analytical Method

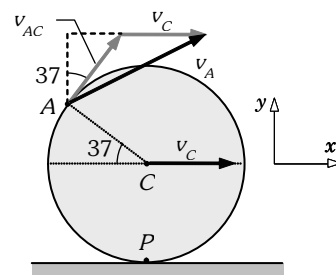
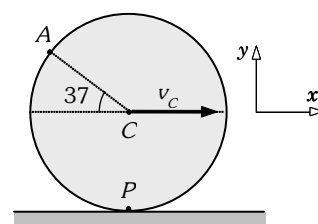
$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \overline{AC} \rightarrow \vec{v}_A = 25\hat{i} + (-5\hat{k}) \times (-5\cos 37^\circ\hat{i} + 5\sin 37^\circ\hat{j})$$

$$\vec{v}_A = (40\hat{i} + 20\hat{j}) \text{ m/s}$$

Superposition Method

In rolling $v_C = v_{AC} = \omega R = 25 \text{ m/s}$. The superposition i.e. vector addition of the terms of equation $\vec{v}_A = \vec{v}_C + \vec{v}_{AC}$ are shown in the following figure. Resolving $v_{AC} = \omega R = 25 \text{ m/s}$ into its Cartesian components and adding to $\vec{v}_C$, we obtain

$$\vec{v}_A = \vec{v}_C + \vec{v}_{A/C} \rightarrow \quad \vec{v}_A = 25\hat{i} + 15\hat{i} + 20\hat{j} = (40\hat{i} + 20\hat{j}) \text{ m/s}$$



Acceleration relations by Analytical Method

The point of contact P does not slide on the surface, therefore component of its acceleration parallel to the surface must be zero. However, it has an acceleration component towards the center. The center always moves parallel to the horizontal surface and does not change direction of its velocity; therefore, its acceleration can only be parallel to the surface.

Relation between acceleration of acceleration vector of the center C and point of contact P can be obtained with the help of relative motion [15] equation together with the above fact.

$$\vec{a}_C = \vec{a}_P + \vec{\alpha} \times \vec{PC} - \omega^2 \vec{PC} \rightarrow a_C \hat{i} = a_{Py} \hat{j} + (-\alpha \hat{k}) \times R \hat{j} - \omega^2 R \hat{j} = a_{Py} \hat{j} + \alpha R \hat{i} - \omega^2 R \hat{j}$$

Equating coefficients of x and y -components on both the sides of the above equation, we have

$$\vec{a}_C = \alpha R \hat{i} \quad [20]$$

$$\vec{a}_P = \omega^2 R \hat{j} \quad [21]$$

The eq. [20] is used as condition for rolling without slipping together with eq. [16]

In the given figure, acceleration vectors the point of contact; center and the top point are shown. Now we will see how these accelerations can be calculated by using relative motion equation.

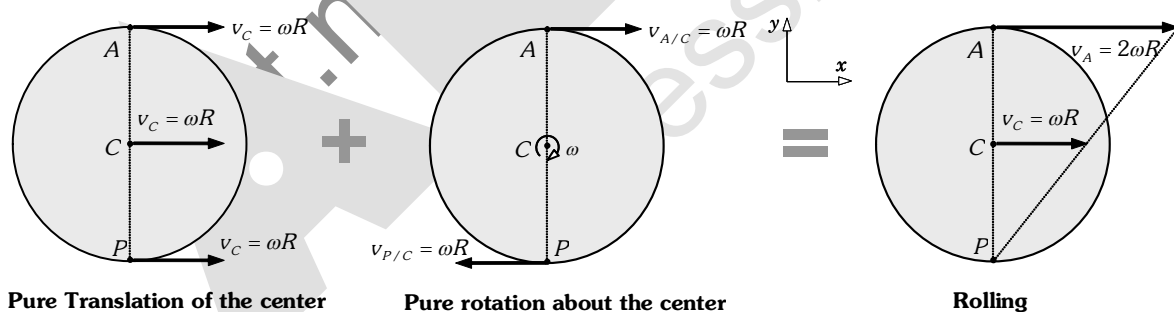
Once velocity of the center is obtained, we can use relative motion between A and C as well. Now we calculate acceleration of the top point A .

$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \vec{CA} - \omega^2 \vec{CA} \rightarrow \vec{a}_A = \alpha R \hat{i} + (-\alpha \hat{k}) \times R \hat{i} - \omega^2 R \hat{i} \\ \vec{a}_A = 2\alpha R \hat{i} - \omega^2 R \hat{i} \quad [22]$$

Acceleration vector of point A and its components are shown in the given figure.

Acceleration relations by Superposition Method

Now we see how acceleration relations are expressed for a rolling wheel by assuming its rolling as superposition of its translation with the velocity of center and simultaneous rotation about the centre.

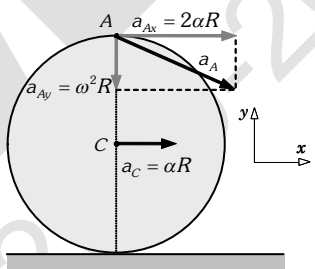
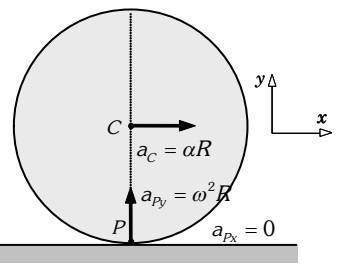


Ex. A body of round section of radius 10 cm starts rolling on a horizontal stationary surface with uniform angular acceleration 2 rad/s^2 .

- Find initial acceleration of the center C and top point A .
- Find expression for acceleration of the top point A as function of time.

Sol. Initially when the body starts, it has no angular velocity; therefore, the last term in relative motion equation [15] for acceleration vanishes and for a pair of two points A and B the equation reduces to

$$\vec{a}_A = \vec{a}_B + \vec{\alpha} \times \vec{BA}$$



The angular acceleration vector is $\vec{\alpha} = -2\hat{k} \text{ rad/s}^2$.

- (a) Acceleration of the center C is obtained by using condition for rolling without slipping.

$$\vec{a}_C = \vec{\alpha} \times \overline{PC} \rightarrow \vec{a}_C = -2\hat{k} \times 10\hat{j} = 20\hat{i} \text{ cm/s}^2 \text{ Ans.}$$

Acceleration of the point A can be obtained either by analytical method, superposition method or by use of ICR. These methods for calculation of acceleration of the top point are already described; therefore, we use the result.

$$\vec{a}_A = 2\alpha R\hat{i} \rightarrow \vec{a}_A = 40\hat{i} \text{ cm/s}^2 \text{ Ans.}$$

- (b) Initially at the instant $t = 0$, when the body starts, its angular velocity is zero. At latter time it acquires angular velocity $\vec{\omega}$, therefore acceleration of any point on the body, other than its center, has an additional component of acceleration. This additional component is accounted for by the last term in the relative motion equation [15].

Angular velocity acquired by the body at time t is obtained by eq.[4] used for a body rotating with constant angular acceleration.

$$\vec{\omega} = \vec{\omega}_0 + \vec{\alpha}t \rightarrow \text{Substituting } \omega_0 = 0, \text{ we have}$$

$$\vec{\omega} = -2t\hat{k}$$

Analytical Method

Using the relative motion equation for the pair of points C and A , we have

$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \overline{CA} - \omega^2 \overline{CA} \rightarrow \vec{a}_A = \alpha R\hat{i} + (-\alpha\hat{k}) \times R\hat{j} - \omega^2 R\hat{j} = 2\alpha R\hat{i} - \omega^2 R\hat{j}$$

Substituting the known values

$$\vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm}$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2 \text{ Ans.}$$

Superposition Method

We superimpose translation motion of the center and rotation motion about the center. In fact it is vector addition of terms of above equation used in analytical method.

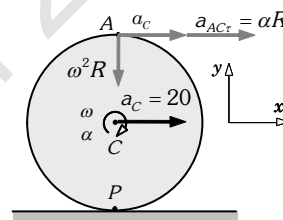
From the above figure, we have

$$\vec{a}_A = (a_C + \alpha R)\hat{i} - \omega^2 R\hat{j}$$

Substituting known values

$$\vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm,}$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2 \text{ Ans.}$$



Methods of Impulse and Momentum

Methods of impulse and momentum describe what happens over a time interval. When motion of a body involves rotation we have to consider angular impulse as well as angular momentum. In this section we discuss concept of angular impulse, angular momentum of rigid body, angular impulse momentum principle and conservation of angular momentum.

Angular Impulse

Like impulse of a force angular impulse of a constant torque equals to product of the torque and concerned time interval and if the torque is not constant it must be integrated with time over the concerned time interval.

If torque $\vec{\tau}_0$ about an axis passing through O is constant, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{0,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{0,1 \rightarrow 2} = \vec{\tau}_0 (t_2 - t_1)$$

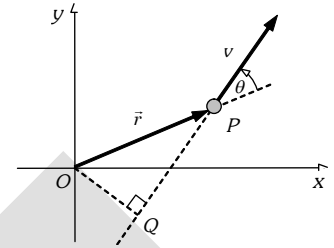
If torque $\vec{\tau}_o$ about an axis passing through O is time varying, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{o,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{o,1 \rightarrow 2} = \int_{t_1}^{t_2} \vec{\tau}_o dt$$

Angular momentum of a particle

Angular momentum $\vec{L}_o$ about the origin O of a particle of mass m moving with velocity $\vec{v}$ is defined as the moment of its linear momentum $\vec{p} = m\vec{v}$ about the point O .

$$\vec{L}_o = \vec{r} \times (m\vec{v})$$



Angular Momentum of a Rigid Body

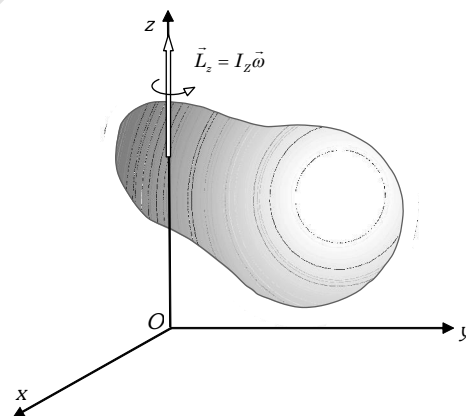
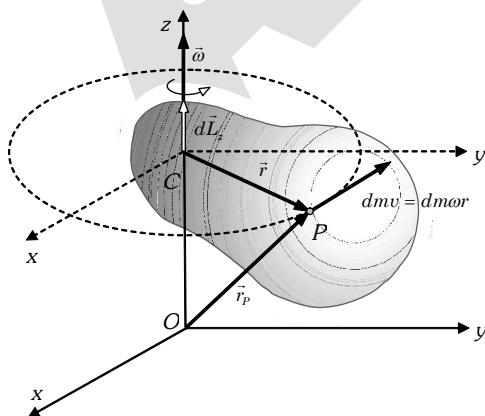
Angular momentum is quantity of rotation motion in a body. The angular momentum of a system of particles is the sum of angular momentum all the particles within the system. A rigid body is an assemblage of large number of particles maintaining their mutual distances intact under all circumstances, therefore angular momentum of a rigid body must be sum of angular momenta of all of its particles.

Angular Momentum about a point and about an axis

Angular momentum of a particle is not defined about an axis instead it is defined about a point. Therefore above idea of summing up angular momenta of all the particles about a point gives angular momentum of the rigid body about a point. But while dealing with fixed axis rotation or rotation about axis in translation we need angular momentum about an axis.

Angular momentum about an axis is calculated similar to torque about an axis. To calculate angular momentum of a particle of rigid body about an axis we take moment of momentum of the particle about the point where plane of motion of the point of application of the force intersects the axis.

In the following figure is shown angular momentum $d\vec{L}_z = \vec{r} \times (dm\vec{v}) = r^2 dm\vec{\omega}$ of a particle P of a rigid body rotating about the z -axis. It is along the z -axis i.e. axis of rotation. In the next figure total angular momentum $\vec{L}_z = \int d\vec{L}_z = I_z \vec{\omega}$ about the axis of rotation is shown. It is also along the axis of rotation.



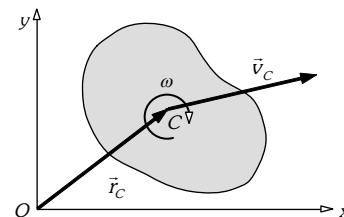
Angular Momentum in general plane motion

Angular momentum of a body in plane motion can also be written similar to torque equation or kinetic energy as sum of angular momentum about the axis due to translation of mass center and angular momentum of centroidal rotation about centroidal axis parallel to the original axis.

Consider a rigid body of mass M in plane motion. At the instant shown its mass center has velocity $\vec{v}_C$ and it is rotating with angular velocity $\vec{\omega}$ about an axis perpendicular to the plane of the figure. Its angular momentum $\vec{L}_O$ about an axis passing through the origin and parallel to the original is expressed by the following equation.

$$\vec{L}_O = \vec{r}_C \times (M\vec{v}_C) + I_C \vec{\omega}$$

The first term of the above equation represent angular momentum due to translation of the mass center and the second term represents angular momentum in centroidal rotation.

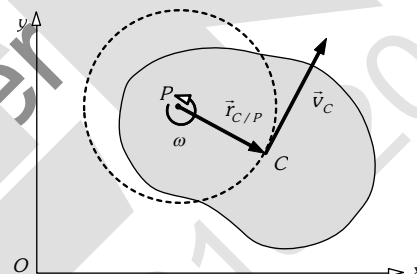


Angular momentum in rotation about fixed axis

Consider a body of mass M rotating with angular velocity ω about a fixed axis perpendicular to plane of the figure passing through point P . Making use of the parallel axis theorem $I_P = Mr_{C/P}^2 + I_C$ and equation $\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P}$ we can express the angular momentum $\vec{L}_P$ of the body about the fixed rotational axis.

$$\vec{L}_P = I_P \vec{\omega}$$

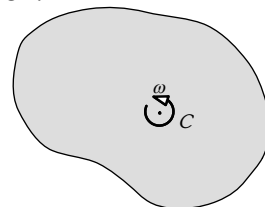
The above equation reveals that the angular momentum of a rigid body in plane motion can also be expressed in a single term due to rotation about the instantaneous axis of rotation.



Angular momentum in pure centroidal rotation

In pure centroidal rotation, mass center remains at rest, therefore angular momentum due to translation of the mass center vanishes.

$$\vec{L}_C = I_C \vec{\omega}$$



Rotational Equivalent of the Newton's Laws of Motion

Differentiating terms on both the sides of equation $\vec{L}_O = \vec{r}_C \times (M\vec{v}_C) + I_C \vec{\omega}$ with respect to time, and making substitution of $\vec{v}_C = d\vec{r}_C/dt$, $\vec{a}_C = d\vec{v}_C/dt$ and $\vec{\alpha} = d\vec{\omega}/dt$ we have

$$\frac{d\vec{L}_O}{dt} = \vec{v}_C \times (M\vec{v}_C) + \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

The first on the right hand side vanishes, so we can write

$$\frac{d\vec{L}_O}{dt} = \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

Now comparing the above equation with torque equation $\Sigma \vec{\tau}_O = \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$, we have

$$\Sigma \vec{\tau}_O = \frac{d\vec{L}_O}{dt}$$

The above equation though developed for plane motion only yet is valid for rotation about an axis in rotation also. It states that the net torque about the origin of an inertial frame equals to the time rate of change in angular momentum about the origin and can be treated as a parallel to Newton's second law which states that net external force on a body equals to time rate of change in its linear momentum.

Angular Impulse Momentum Principle

Rearranging the terms and integrating both the sides obtained from previous equation, we can write

$$\Sigma \int_{t_1}^{t_2} \vec{\tau}_o dt = \vec{L}_{o2} - \vec{L}_{o1}$$

The left hand side of the above equation is the angular impulse of torque of all the external forces in the time interval in the time interval t_1 to t_2 .

$$\Sigma \vec{J}_{o,1 \rightarrow 2} = \vec{L}_{o2} - \vec{L}_{o1}$$

The idea expressed by the above equation is known as angular impulse momentum principle and states that increment in the angular momentum of a body about a point in a time interval equals to the net angular impulse of all the external forces acting on it during the concerned time interval.

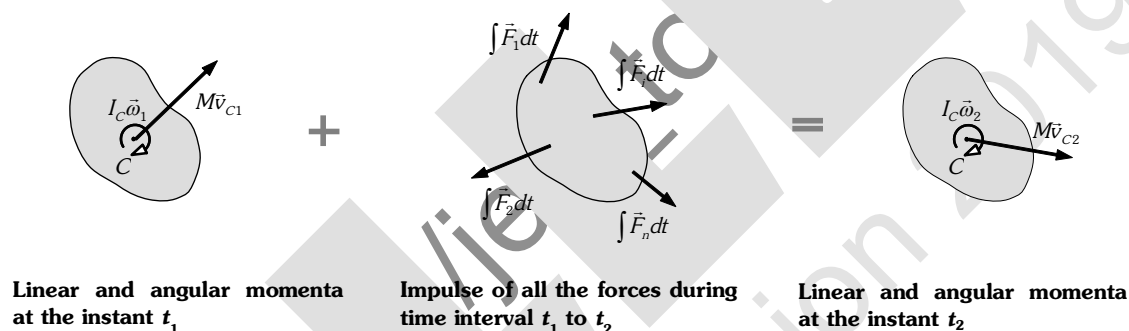
For the ease of application the above equation is rearranged as

$$\vec{L}_{o1} + \Sigma \vec{J}_{o,1 \rightarrow 2} = \vec{L}_{o2}$$

Like linear impulse momentum principle, the angular impulse momentum principle provides us solution of problems concerned with change in angular velocity in a time interval or change in angular velocity during very short interval interactions.

Method of Impulse Momentum Principle for Plane motion of a Rigid Body

Linear momentum and angular momentum serve as measures of amount of translation and rotation motion respectively. The external forces acting on a rigid body can change its state of translation as well as rotation motion which is reflected by change in linear as well as angular momentum according to the principles of linear impulse and momentum and angular impulse and momentum.



In the above figure is shown strategy to apply method of impulse and momentum. Consider a rigid body of mass M in plane motion. Its moment of inertia about the centroidal axis perpendicular to plane of motion is I_C . Let $\vec{v}_{c1}$ and $\vec{\omega}_1$ represent velocity of its mass center and its angular velocity at the beginning of a time interval t_1 to t_2 . Under the action of several forces $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_i, \dots, \vec{F}_n$ during the time interval its mass center velocity and angular velocity become $\vec{v}_{c2}$ and $\vec{\omega}_2$ respectively. The adjacent figure shows strategy representing how to write equations for linear and angular impulse momentum principles.

While applying the principle it becomes simpler to consider translation of the mass center and centroidal rotation separately. Thus in an alternative way we apply linear impulse momentum principle for translation of the mass center and angular impulse momentum principle for centroidal rotation.

Translation of mass center:

Linear impulse momentum principle.

$$\vec{p}_1 + \Sigma \vec{I}_{mp1 \rightarrow 2} = \vec{p}_2$$

Here $\vec{p}_1 = M\vec{v}_{c1}$ and $\vec{p}_2 = M\vec{v}_{c2}$ represent linear momentums at the beginning and end of the time interval and $\Sigma \vec{I}_{mp1 \rightarrow 2}$ stands for impulse of all the external forces during the time interval.

Centroidal rotation:

Angular impulse momentum principle.

$$\vec{L}_{c1} + \Sigma \vec{J}_{c,1 \rightarrow 2} = \vec{L}_{c2}$$

Here $\vec{L}_{C1} = I_C \vec{\omega}_1$ and $\vec{L}_{C2} = I_C \vec{\omega}_2$ represent angular momentums about the centroidal axis at the beginning and end of the time interval and $\sum \vec{J}_{C,1 \rightarrow 2}$ stands for angular impulse of all the external forces about the centroidal axis during the time interval.

Conservation of Angular Momentum

If angular impulse of all the external forces about an axis in time interval vanishes, the angular momentum of the system about the same axis in that time interval remain unchanged.

$$\text{If } \sum \int_{t_1}^{t_2} \vec{\tau}_o dt = 0, \text{ we have } \vec{L}_{o1} = \vec{L}_{o2}$$

The condition of zero net angular impulse required for conservation of angular momentum can be fulfilled in the following cases.

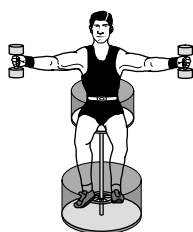
- If no external force acts, the angular impulse about all axes will be zero and hence angular momentum remains conserved about all axes.
- If net torque of all the external forces or torques of each individual force about an axis vanishes the angular momentum about that axes will be conserved.
- If all the external forces are finite in magnitude and the concerned time interval is infinitely small, the angular momentum remain conserved.
- If a system of rigid bodies changes its moment of inertia by changing its configuration due to internal forces only its angular momentum about any axes remains conserved. If we denote the moment of inertias in two configurations by I_1 and I_2 and angular velocities by ω_1 and ω_2 , we can write

$$I_1 \vec{\omega}_1 = I_2 \vec{\omega}_2$$

The principle of conservation of angular momentum governs a wide range of physical processes from subatomic to celestial world. The following examples explicate some of these applications.

Student on rotating turntable

The student, the turntable and dumbbells make an isolated system on which no external torque acts, if we ignore friction in the bearing of the turntable and air friction. Initially the student has his arm stretched on rotating turntable. When he pulls dumbbells close to his body, angular velocity increases due to conservation of angular momentum.

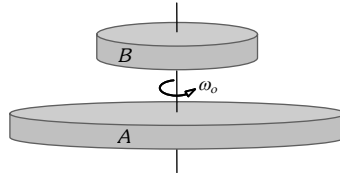


Larger moment of inertia and smaller angular velocity



Smaller moment of inertia and larger angular velocity

Ex. Consider the disk A of moment of inertia I_1 rotating freely in horizontal plane about its axis of symmetry with angular velocity ω_0 . Another disk B of moment of inertia I_2 held at rest above the disk A . The axis of symmetry of the disk B coincides with that of the disk A as shown in the figure. The disk B is released to land on the disk A . When sliding stops, what will be the angular velocity of both the disks?



Sol. Both the disks are symmetric about the axis of rotation therefore does not require any external torque to keep the axis stationary. When the disk B lands on A slipping starts. The force of friction provides an internal torque to system of both the disk. It slows down rotation rate of A and increases that of B till both acquire same angular velocity ω .

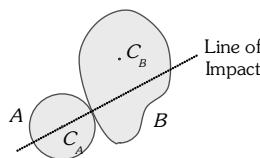
Since there is no external torques on the system of both the disks about the axis of rotation, the total angular momentum of the system remains conserved. The total angular momentum of the system is the sum of angular momentum of both disks. Denoting the angular momentum of the disk A before B lands on it and long after slipping between them stops by symbols $\vec{L}_{A1}$, $\vec{L}_{B1}$, $\vec{L}_{A2}$ and $\vec{L}_{B2}$ respectively, we can express conservation of angular momentum by the following equation.

$$\vec{L}_{A1} + \vec{L}_{B1} = \vec{L}_{A2} + \vec{L}_{B2} \rightarrow I_1\omega_0 + 0 = I_1\omega + I_2\omega \Rightarrow \omega = \frac{I_1\omega_0}{I_1 + I_2} \quad \text{Ans.}$$

Eccentric Impact

In eccentric impact the line of impact which is the common normal drawn at the point of impact does not pass through mass center of at least one of the colliding bodies. It involves change in state of rotation motion of either or both the bodies.

Consider impact of two A and B such that the mass center C_B of B does not lie on the line of impact as shown in figure. If we assume bodies to be frictionless their mutual forces must act along the line of impact. The reaction force of A on B does not pass through the mass center of B as a result state of rotation motion of B changes during the impact.



Problems of Eccentric Impact

Problems of eccentric impact can be divided into two categories. In one category both the bodies under going eccentric impact are free to move. No external force act on either of them. Their mutual forces are responsible for change in their momentum and angular momentum. In another category either or both of the bodies are hinged.

Eccentric Impact of bodies free to move

Since no external force acts on the two body system, we can use principle of conservation of linear momentum, principle of conservation of angular momentum about any point and concept of coefficient of restitution.

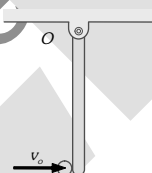
The coefficient of restitution is defined for components of velocities of points of contacts of the bodies along the line of impact.

While applying principle of conservation of angular momentum care must be taken in selecting the point about which we write the equation. The point about which we write angular momentum must be at rest relative to the selected inertial reference frame and as far as possible its location should be selected on line of velocity of the mass center in order to make zero the first term involving moment of momentum of mass center.

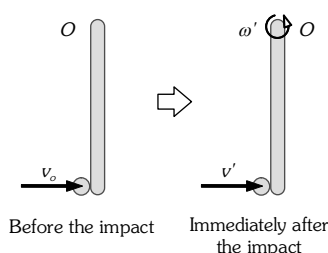
Eccentric Impact of hinged bodies

When either or both of the bodies are hinged the reaction of the hinge during the impact act as external force on the two body system, therefore linear momentum no longer remain conserved and we cannot apply principle of conservation of linear momentum. When both the bodies are hinged we cannot also apply conservation of angular momentum, and we have to use impulse momentum principle on both the bodies separately in addition to making use of coefficient of restitution. But when one of the bodies is hinged and other one is free to move, we can apply conservation of angular momentum about the hinge.

- Ex.** A uniform rod of mass m and length ℓ is suspended from a fixed support and can rotate freely in the vertical plane. A small ball of mass m moving horizontally with velocity v_0 strikes elastically the lower end of the rod as shown in the figure. Find the angular velocity of the rod and velocity of the ball immediately after the impact.



- Sol.** The rod is hinged and the ball is free to move. External forces acting on the rod ball system are their weights and reaction from the hinge. Weight of the ball as well as the rod are finite and contribute negligible impulse during the impact, but impulse of reaction of the hinge during impact is considerable and cannot be neglected. Obviously linear momentum of the system is not conserved. The angular impulse of the reaction of hinge about the hinge is zero. Therefore angular momentum of the system about the hinge is conserved. Let velocity of the ball after the impact becomes v'_B and angular velocity of the rod becomes ω' .



We denote angular momentum of the ball and the rod about the hinge before the impact by L_{B1} and L_{R1} and after the impact by L_{B2} and L_{R2} .

Applying conservation of angular momentum about the hinge, we have

$$\vec{L}_{B1} + \vec{L}_{R1} = \vec{L}_{B2} + \vec{L}_{R2} \rightarrow mv_o \ell + 0 = mv'_B \ell + I_o \omega'$$

Substituting $\frac{1}{3}M\ell^2$ for I_o , we have

$$3mv'_B + M\ell\omega' = 3mv_o \quad (1)$$

The velocity of the lower end of the rod before the impact was zero and immediately after the impact it becomes $\ell\omega'$ towards right. Employing these facts we can express the coefficient of restitution according to eq.

$$e = \frac{v'_{Qn} - v'_{Pn}}{v_{pn} - v_{Qn}} \rightarrow \ell\omega' - v'_B = ev_o \quad (2)$$

From eq. (1) and (2), we have

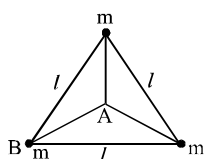
Velocity of the ball immediately after the impact $v'_B = \frac{(3m - eM)v_o}{3m + M}$ **Ans.**

Angular velocity of the rod immediately after the impact $\omega' = \frac{3(1+e)mv_o}{(3m+M)\ell}$ **Ans.**

EXERCISE (S-1)

Moment of inertia

1. Three equal masses m are rigidly connected to each other by massless rods of length l forming an equilateral triangle, as shown in the figure. What is the ratio of the moment of inertia of the assembly for an axis through B compared with that for an axis through A (centroid). Both the axis are perpendicular to the plane of triangle.



2. Two rods of equal mass m and length l lie along the x axis and y axis with their centres at origin. What is the moment of inertia of the system about the line $x = y$:

Torque and its calculation

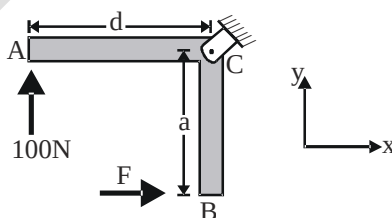
3. A force $\vec{F} = 2\hat{i} + 3\hat{j}$ N is applied to an object that is pivoted about a fixed axle aligned along the z coordinate axis. If the force is applied at the point $\vec{r} = 4\hat{i} + 5\hat{j}$ m, find (a) the magnitude of the net torque about the z axis and (b) the direction of the torque vector τ .

Equilibrium

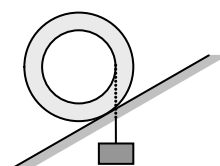
4. In an experiment with a massless beam balance an unknown mass m is balanced by two known masses of 16kg and 4 kg as shown in figure. Find the value of the unknown mass m .



5. Find the force F required to keep the system in equilibrium. The dimensions of the system are $d = 0.3 \text{ m}$ and $a = 0.2 \text{ m}$. Assume the rods to be massless.

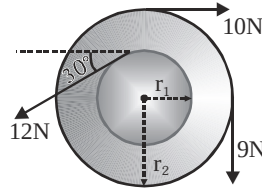


6. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline?

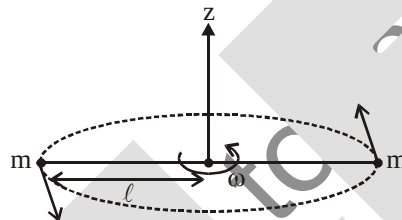


$\tau = I\alpha$ and its calculation

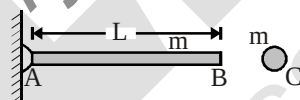
7. In the following figure r_1 and r_2 are 5 cm and 30 cm respectively. If the moment of inertia of the wheel is 5100 kg-m^2 about the axis passing through its centre and perpendicular to the plane of wheel, then what will be its angular acceleration?



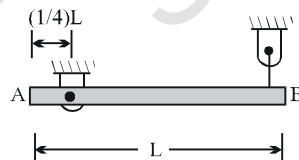
8. A dumbbell consists of two identical particles of mass m connected by a rigid light rod of length 2ℓ . The dumbbell is set spinning with angular speed ω_0 on a surface with a small friction coefficient μ_k . If dumbbell stops in time $t = \frac{K\omega_0\ell}{2\mu g}$ where K is a constant, then find the value of K .



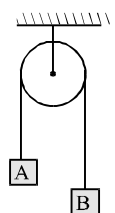
9. A uniform bar AB of mass m and a ball of the same mass are released from rest from the same horizontal position. The bar is hinged at end A. There is gravity downwards. What is the distance of the point from point B that has the same acceleration as that of ball, immediately after release?



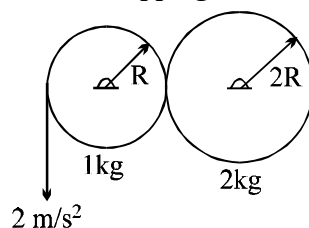
10. A uniform beam of length L and mass m is supported as shown. If the cable at B suddenly breaks, determine; (a) the acceleration of end B. (b) the reaction at the pin support.



11. In the figure, A & B are two blocks of mass 4 kg & 2 kg respectively attached to the two ends of a light string passing over a disc C of mass 40 kg and radius 0.1 m. The disc is free to rotate about a fixed horizontal axes, coinciding with its own axis. The system is released from rest and the string does not slip over the disc. Find:
- the linear acceleration of mass B.
 - the number of revolutions made by the disc at the end of 10sec. from the start.
 - the tension in the string segment supporting the block A.

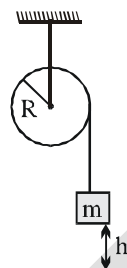


12. Two discs A and B touch each other as shown in figure. A rope tightly wound on A is pulled down at 2 m/s^2 . Find the friction force between A and B if slipping is absent

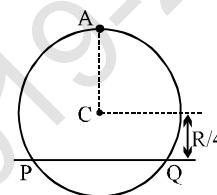


Kinetic energy in pure rotation

13. A mass m is attached to a pulley through a cord as shown in the figure. The pulley is a solid disk with radius R . The cord does not slip on the disk. The mass is released from rest at a height h from the ground and at the instant the mass reaches the ground, the disk is rotating with angular velocity ω . Find the mass of the disk.



14. A uniform circular disc has radius R and mass m . A particle also of mass m is fixed at a point A on the edge of the disc as in figure. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance $R/4$ from the centre C of the disc. The line AC is perpendicular to PQ. Initially the disc is held vertical with the point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle at it reaches its lowest position.

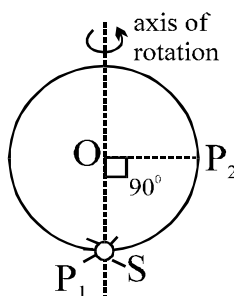


15. A rod of mass m & length ℓ is hinged at its upper end. It can rotate in vertical plane. It is given angular velocity ω so that it can complete vertical circle. Find (a) ω (b) Tension at centre of rod at initial moment.

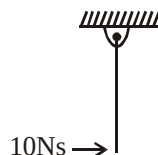


Angular momentum and its conservation

16. A uniform ring is rotating about vertical axis with angular velocity ω initially. A point insect (S) having the same mass as that of the ring starts walking from the lowest point P_1 and finally reaches the point P_2 (as shown in figure). What is the final angular velocity of the ring?

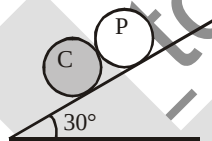


17. Two men, each of mass 75 kg, stand on the rim of a horizontal large disc, diametrically opposite to each other. The disc has a mass 450 kg and is free to rotate about its axis. Each man simultaneously start along the rim clockwise with the same speed and reaches their original starting points on the disc. Find the angle turned by the disc with respect to the ground in this duration.
18. A thin uniform straight rod of mass 2 kg and length 1 m is free to rotate about its upper end when at rest. It receives an impulsive blow of 10 Ns at its lowest point, normal to its length as shown in figure. Find the kinetic energy of rod just after impact.

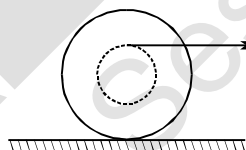


Combined rotation and translation

19. A solid uniform disk of mass m rolls down a fixed inclined plane without slipping with an acceleration a . Find the frictional force on the disk due to surface of the plane :
20. A solid cylinder C and a hollow pipe P of same diameter are in contact when they are released from rest as shown in the figure on a long incline plane. Cylinder C and pipe P roll without slipping. Determine the clear gap (in m) between them after 6 seconds.

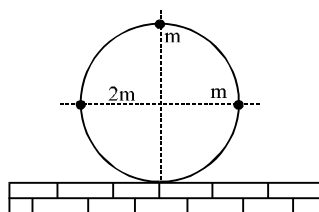


21. A spool of inner radius R and outer radius $3R$ has a moment of inertia $= MR^2$ about an axis passing through its geometric centre, where M is the mass of the spool. A thread wound on the inner surface of the spool is pulled horizontally with a constant force $= Mg$. Find the acceleration of the point on the thread which is being pulled assuming that the spool rolls purely on the floor.

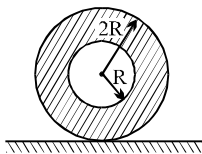


Kinetic energy in rolling

22. A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0 . Find the kinetic energy of the system. (Slipping is absent)

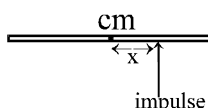


23. A hollow cylinder with inner radius R , outer radius $2R$ and mass M is rolling with speed of its axis v . Its kinetic energy is :-

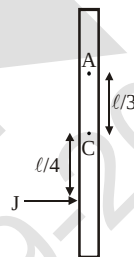


Angular impulse

24. A uniform rod of length l is given an impulse at right angle to its length as shown. Find the distance of instantaneous centre of rotation from the centre of the rod.



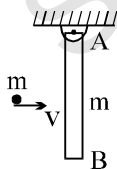
25. A uniform rod of mass m and length ℓ is placed in gravity free space and linear impulse J is given to the rod at a distance $x = \ell/4$ from centre 'C' and perpendicular to the rod. Point A is at a distance $\ell/3$ from centre as shown in the figure. Then find
- Speed of centre of rod
 - Speed of point A
 - Speed of upper end of rod
 - Speed of lower end of rod



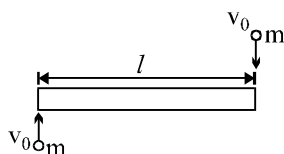
26. A solid sphere of mass m and radius R is placed on a smooth horizontal surface. A sudden blow is given horizontally to the sphere at a height $h = 4R/5$ above the centre line. If I is the impulse of the blow then find
- the minimum time after which the highest point B will touch the ground
 - the displacement of the centre of mass during this interval.

Eccentric collision

27. A uniform rod AB of length L and mass m is suspended freely at A and hangs vertically at rest when a particle of same mass m is fired horizontally with speed v to strike the rod at its mid point. If the particle comes to rest after the impact, then find the impulsive reaction at A.

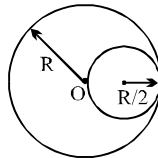


28. On a smooth table two particles of mass m each, travelling with a velocity v_0 in opposite directions, strike the ends of a rigid massless rod of length l , kept perpendicular to their velocity. The particles stick to the rod after the collision. Find the tension in rod during subsequent motion.

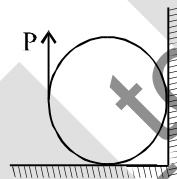


EXERCISE (S-2)

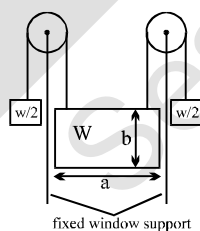
1. A hole of radius $R/2$ is cut from a solid sphere of radius R . If the mass of the remaining part of sphere is M , then find moment of inertia of the body about an axis through O .



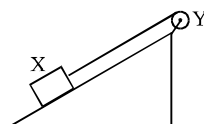
2. Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ .
3. Figure shows a vertical force applied tangentially to a uniform cylinder of weight F_g . The coefficient of static friction between the cylinder and both surfaces is 0.500 . In terms of F_g , find the maximum force P that can be applied that does not cause the cylinder to rotate.



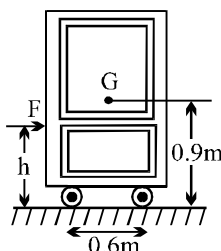
4. A slight loosely fit window is balanced by two strings which are connected to weights $w/2$ each. The strings pass over the frictionless pulleys as shown in the figure. The strings are tied almost at the corner of the window. The string on the right is cut and then the window accelerates downwards. If the coefficients of friction between the window and the side supports is μ then calculate the acceleration of the window in terms of μ , a , b and g , where a is width and b is the length of the window.



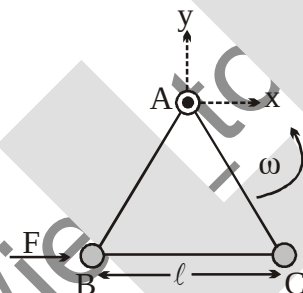
5. A block X of mass 0.5kg is held by a long massless string on a frictionless fixed inclined plane of inclination 30° to the horizontal. The string is wound on a uniform solid cylindrical drum Y of mass 2kg and of radius 0.2m as shown in the fig. The drum is given an initial angular velocity such that the block X starts moving up the plane. ($g = 9.8\text{ m/s}^2$)
- (i) Find the tension in the string during the motion
- (ii) At a certain instant of time the magnitude of the angular velocity of Y is 10 rad/sec . Calculate the distance travelled by X from that instant of time until it comes to rest.



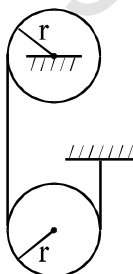
6. A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown, determine
- the acceleration of the cabinet,
 - the range of values of h for which the cabinet will not tip.



7. Three particles A, B, and C each of mass m , are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side ℓ . This body is placed on a horizontal frictionless table (x - y plane) and is hinged to it at the point A, so that it can move without friction about the vertical axis through A (see figure). The body is set into rotational motion on the table about A with a constant angular velocity ω .
- Find the magnitude of the horizontal force exerted by the hinge on the body.
 - At time T , when the side BC is parallel to the x -axis, a force F is applied on B along BC (as shown). Obtain the x -component and the y -component of the force exerted by the hinge on the body, immediately after time T .
- [IIT-JEE' 2001]

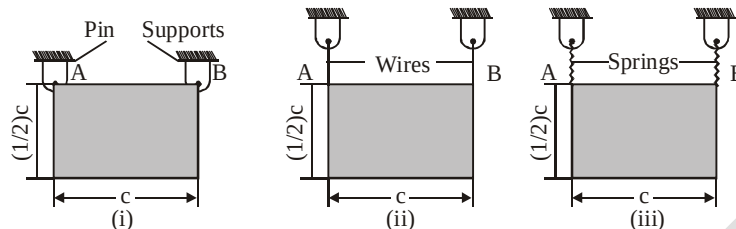


8. The door of an automobile is open and perpendicular to the body. The automobile starts with an acceleration of 2 ft/sec^2 , and the width of the door is 30 inches. Treat the door as a uniform rectangle, and neglect friction to find the speed of its outside edge as seen by the driver when the door closes.
9. Two uniform cylinders, each of mass $m = 10 \text{ kg}$ and radius $r = 150 \text{ mm}$, are connected by a rough belt as shown. If the system is released from rest, determine
- the velocity of the centre of cylinder A after it has moved through 1.2 m &
 - the tension in the portion of the belt connecting the two cylinders.

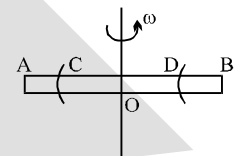


10. A circular disc of mass 300 gm and radius 20 cm can rotate freely about a vertical axis passing through its centre O. A small insect of mass 100 gm is initially at a point A on the disc (which is initially stationary) the insect starts walking from rest along the rim of the disc with such a time varying relative velocity that the disc rotates in the opposite direction with a constant angular acceleration $= 2\pi \text{ rad/s}^2$. After some time T , the insect is back at the point A. By what angle θ has the disc rotated till now ; as seen by a stationary earth observer? Also find the time T .

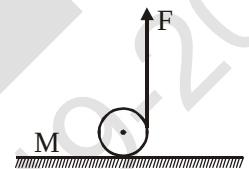
11. A uniform plate of mass m is suspended in each of the ways shown. For each case determine immediately after the connection at B has been released ;
- the angular acceleration of the plate .
 - the acceleration of its mass center .



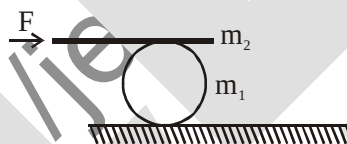
12. A rigid horizontal smooth rod AB of mass 0.75 kg and length 40 cm can rotate freely about a fixed vertical axis through its mid point O. Two rings each of mass 1 kg are initially at rest at a distance of 10 cm from O on either side of the rod. The rod is set in rotation with an angular velocity of 30 radians per second. Find the velocity of each ring along the length of the rod in m/s when they reach the ends of the rod



13. The free end of the string wound on the surface of a solid cylinder of mass M & radius r is pulled up by a force F as shown. If there is sufficient friction between cylinder & floor so that the cylinder is able to roll without slipping, find the maximum angular acceleration that the cylinder can have (without cylinder leaving contact with surface).



14. A man pushes a cylinder of mass m_1 with help of a plank of mass m_2 as shown. There is no slipping at any contact. the horizontal component of the force applied by the man is F . Find:



- the accelerations of the plank and the centre of mass of the cylinder, and
- the magnitudes and directions of frictional forces at contact points. [JEE '99, 6 + 4]

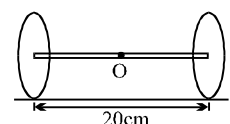
15. A 160 mm diameter pipe of mass 6 kg rests on a 1.5 kg plate. The pipe and plate are initially at rest when a force P of magnitude 25 N is applied for 0.75s. Knowing that $\mu_s = 0.25$ & $\mu_k = 0.20$ between the plate and both the pipe and the floor, determine;

- whether the pipe slides with respect to the plate .
- the resulting velocities of the pipe and of the plate .

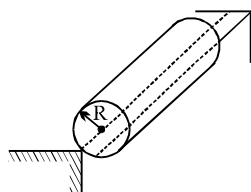


16. Two thin circular disks of mass 2kg and radius 10cm each are joined by a rigid massless rod of length 20cm. The axis of the rod is along the perpendicular to the planes of the disks through their centre. The object is kept on a truck in such a way that the axis of the object is horizontal and perpendicular to the direction of motion of the truck. Its friction with the floor of the truck is large enough so that the object can roll on the truck without slipping. Take x-axis as the direction of motion of the truck and z -axis as the vertically upwards direction. If the truck has an acceleration of 9m/s^2 calculate :

- the force of friction on each disk
- The magnitude and the direction of the frictional torque acting on each disk about the centre of mass O of the object. Express the torque in the vector form of unit vectors in the x-y and z directions.



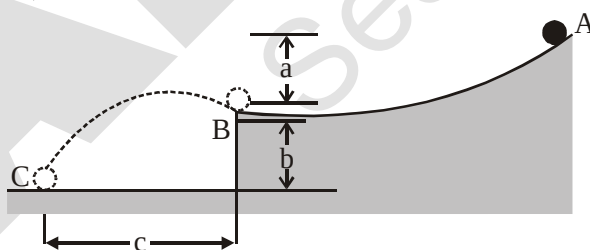
17. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in figure. there is sufficient friction present at the edge so that a very small displacement cause the cylinder to roll of the edge without slipping. Determine
- the angle θ_c through which the cylinder rotates before it leaves contact with the edge,
 - the speed of the centre of mass of the cylinder before leaving contact with the edge, and
 - the ratio of the translational to rotational kinetic energies of the cylinder when its centre of mass is in horizontal line with the edge.



18. A carpet of mass 'M' made of inextensible material is rolled along its length in the form of a cylinder of radius 'R' and is kept on a rough floor. The carpet starts unrolling without sliding on the floor when a negligibly small push is given to it. The horizontal velocity of the axis of the cylindrical part of the carpet when its radius reduces to $R/2$ will be:



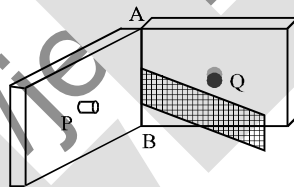
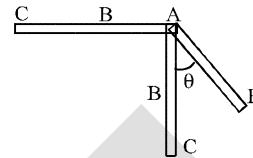
19. A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that $a = 1.5$ m & $b = 1.2$ m, determine;
- the speed of the sphere as it strikes the ground at C .
 - the corresponding distance c .



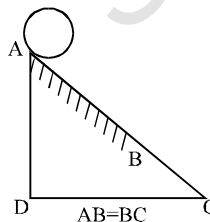
20. A uniform disk of mass m and radius R is projected horizontally with velocity v_0 on a rough horizontal floor so that it starts off with a purely sliding motion at $t = 0$. After t_0 seconds it acquires a purely rolling motion as shown in figure.
- Calculate the velocity of the centre of mass of the disk at t_0 .
 - Assuming the coefficient of friction to be μ calculate t_0 . Also calculate the work done by the frictional force as a function of time and the total work done by it over a time t much longer than t_0 .



21. A uniform disc of mass m and radius R rotates about a fixed vertical axis passing through its centre with angular velocity ω . A particle of same mass m and having velocity $2\omega R$ towards centre of the disc collides with the disc moving horizontally and sticks to its rim. Find
 (a) the angular velocity of the disc.
 (b) the impulse on the particle due to disc.
 (c) the impulse on the disc due to hinge.
22. A rod hinged at one end is released from the horizontal position as shown in the figure. When it becomes vertical its lower half separates without exerting any reaction at the breaking point. Then find the maximum angle ' θ ' made by the hinged upper half with the vertical.
23. Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 Kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 Kg-m^2 . Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB . This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/s and turns back. If the impulse on the sheet due to each obstacle is 6 N-s . [IIT-JEE' 2001]
 (a) Find the location of the centre of mass of the laminar sheet from AB .
 (b) At what angular velocity does the laminar sheet come back after the first impact ?
 (c) After how many impacts, does the laminar sheet come to rest ?



24. Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B . Find the ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C .



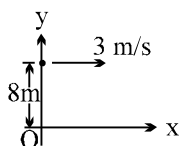
25. A rod AB of mass M and length L is lying on a horizontal frictionless surface. A particle of mass m travelling along the surface hits the end ' A ' of the rod with a velocity v_0 in the direction perpendicular to AB . The collision is completely elastic. After the collision the particle comes to rest.
 (a) Find the ratio m/M . [IIT-JEE' 2000]
 (b) A point P on the rod is at rest immediately after the collision. Find the distance AP .
 (c) Find the linear speed of the point P at a time $\pi L/(3v_0)$ after the collision.

EXERCISE (O-1)

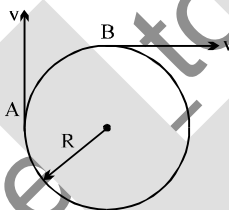
SINGLE CORRECT TYPE QUESTIONS

Concept of ω , α , v , a

1. A particle starts from the point (0m, 8m) and moves with uniform velocity of $3\hat{i}$ m/s. After 5 seconds, the angular velocity of the particle about the origin will be :



- (A) $\frac{8}{289}$ rad/s (B) $\frac{3}{8}$ rad/s (C) $\frac{24}{289}$ rad/s (D) $\frac{8}{17}$ rad/s
2. Two points of a rigid body are moving as shown. The angular velocity of the body is:

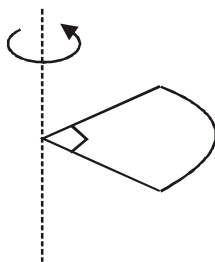


- (A) $\frac{v}{2R}$ (B) $\frac{v}{R}$ (C) $\frac{2v}{R}$ (D) $\frac{2v}{3R}$

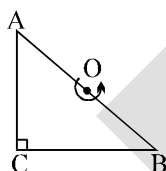
Moment of inertia

3. Three bodies have equal mass m . Body A is solid cylinder of radius R , body B is a square lamina of side R , and body C is a solid sphere of radius R . Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)
- (A) A (B) B (C) C (D) A and C both
4. For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body
- (A) a disc of radius a (B) a ring of radius a
 (C) a square lamina of side $2a$ (D) four rods forming a square of side $2a$
5. Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB. The moment of inertia of the plate about the axis CD is then equal to
- (A) I (B) $I \sin^2\theta$ (C) $I \cos^2\theta$ (D) $I \cos^2(\theta/2)$

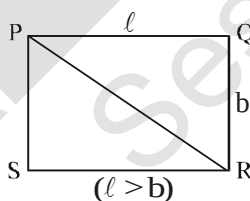
6. One quarter sector is cut from a uniform circular disc of radius R . This sector has mass M . It is made to rotate about a line perpendicular to its plane and passing through the center of the original disc. Its moment of inertia about the axis of rotation is [IIT-JEE 2000]



- (A) $\frac{1}{2} MR^2$ (B) $\frac{1}{4} MR^2$ (C) $\frac{1}{8} MR^2$ (D) $\sqrt{2} MR^2$
7. Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M , about an axis perpendicular to the plane of the plate and passing through the mid point of side AB . (Side $AC = BC = a$)



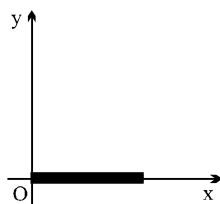
- (A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$ (C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$
8. A circular disc X of radius R is made from an iron plate of thickness t and another disc Y of radius $4R$ is made from an iron plate of thickness $t/4$. Then the relation between the moment of inertia I_X and I_Y is- [AIEEE - 2003]
- (A) $I_Y = 32 I_X$ (B) $I_Y = 16 I_X$ (C) $I_Y = I_X$ (D) $I_Y = 64 I_X$
9. Moment of inertia of a rectangular plate about an axis passing through P and perpendicular to the plate is I . Then moment of PQR about an axis perpendicular to the plane of the plate:



- (A) about $P = I/2$ (B) about $R = I/2$ (C) about $P > I/2$ (D) about $R > I/2$
10. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bent in the form of a semicircular arc. Now its moment of inertia through the centre of the semi circular arc and perpendicular to its plane is I_2 . The ratio of $I_1 : I_2$ will be _____
- (A) < 1 (B) > 1 (C) $= 1$ (D) can't be said
11. One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B such that- [AIEEE - 2004]
- (A) $I_A = I_B$ (B) $I_A > I_B$ (C) $I_A < I_B$ (D) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$

where d_A and d_B are their densities.

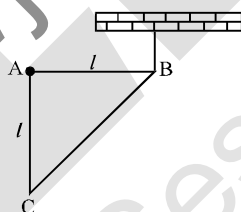
12. The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is :-



- (A) an ellipse (B) a circle (C) a parabola (D) a straight line
- 13.** A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x , the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x-coordinate of centre of mass is
- (A) $x = 2$ (B) $x = 0$ (C) $x = 1$ (D) $x = 3$

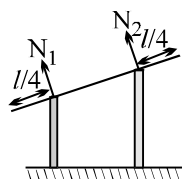
Equilibrium

14. A weightless rod is acted on by upward parallel forces of 2N and 4N at ends A and B respectively. The total length of the rod is $AB = 3\text{m}$. To keep the rod in equilibrium a force of 6N should act in the following manner:
- (A) Downwards at any point between A and B.
 - (B) Downwards at mid point of AB.
 - (C) Downwards at a point C such that $AC = 1\text{m}$.
 - (D) Downwards at a point D such that $BD = 1\text{m}$.
15. A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A in equilibrium is:



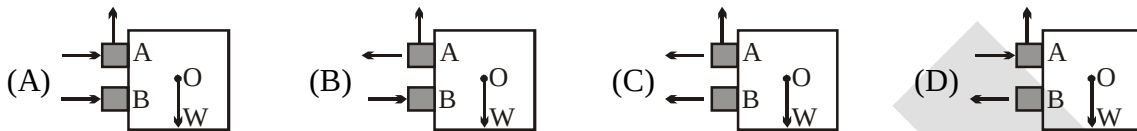
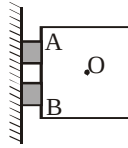
- (A) $\frac{mg}{3}$ (B) $\frac{2mg}{3}$ (C) $\frac{mg}{2}$ (D) mg

- 16.** A uniform rod of length l is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then :-

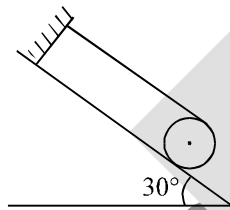


- (A) $N_1 > N_2$
(B) $N_1 < N_2$
(C) $N_1 = N_2$
(D) N_1 and N_2 would be in the vertical directions

17. A vertical rectangular door with its centre of gravity at O (see figure) is fixed on two hinges A and B along one vertical length side of the door. The entire weight of the door is supported by the hinge A. Then the free body force diagram for the door (the arrows indicate the direction of the forces) is :-

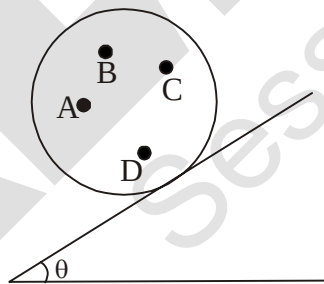


18. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure. The minimum coefficient of friction needed for this configuration to be in equilibrium is:



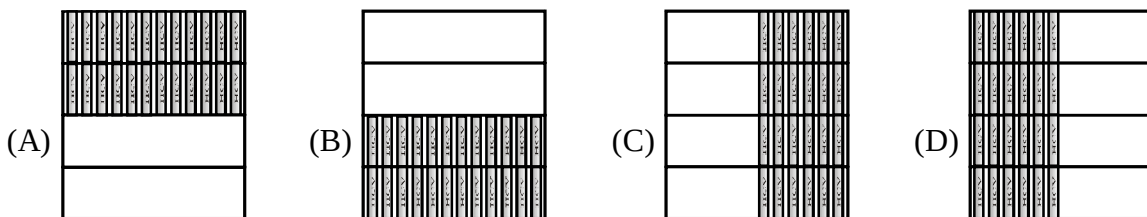
- (A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{2\sqrt{3}}$

19. A non uniform sphere can be kept on a rough inclined plane so that it is in equilibrium. In the figure below the dots represents location of centre of mass. In which one of the positions can sphere be in equilibrium?

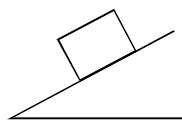


Toppling

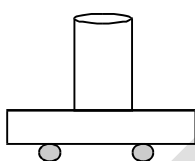
20. Same number of books are placed in four book cases as shown. Which bookcase is most likely to topple forward if pulled a little at the top towards right :



21. A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick which applies greater pressure on the plane is :

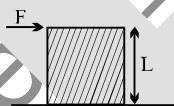


- (A) left half (B) right half
(C) both applies equal pressure (D) the answer depend upon coefficient of friction
22. A uniform 2 kg cylinder rests on a laboratory cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is closest to the maximum acceleration of the cart such that cylinder neither slips nor tips over?



- (A) 2 m/s^2 (B) 4 m/s^2 (C) 5 m/s^2 (D) 6 m/s^2
23. A cubical block of side L rests on a rough horizontal surface with coefficient of friction μ . A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is:

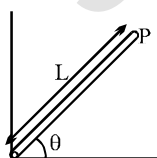
[IIT-JEE'(Scr)'2000]



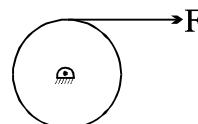
- (A) infinitesimal (B) $mg/4$ (C) $mg/2$ (D) $mg(1-\mu)$

$\tau = I\alpha$

24. A uniform flag pole of length L and mass M is pivoted on the ground with a frictionless hinge. The flag pole makes an angle θ with the horizontal. The moment of inertia of the flag pole about one end is $(1/3)ML^2$. If it starts falling from the position shown in the accompanying figure, the linear acceleration of the free end of the flag pole (labeled P) immediately after it starts falling off would be:

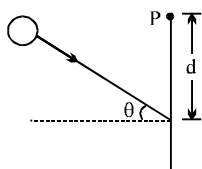


- (A) $(2/3)g\cos\theta$ (B) $(2/3)g$ (C) g (D) $(3/2)g\cos\theta$
25. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As time increases,
- (A) its angular velocity increases, but force on hinge remains constant
(B) its angular velocity remains same, but force on hinge increases
(C) its angular velocity increases and force on hinge increases
(D) its angular velocity remains same and force on hinge is constant

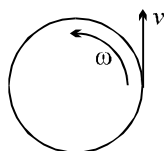


Angular momentum and its conservation

26. A particle of mass 2 kg located at the position $(\hat{i} + \hat{j})$ m has a velocity $2(+\hat{i} - \hat{j} + \hat{k})$ m/s. Its angular momentum about z-axis in $\text{kg-m}^2/\text{s}$ is:
 (A) zero (B) +8 (C) 12 (D) -8
27. A ball of mass m moving with velocity v , collide with the wall elastically as shown in the figure. After impact the change in angular momentum about P is:



- (A) $2 mvd$ (B) $2 mvd \cos \theta$ (C) $2 mvd \sin \theta$ (D) zero
28. Two uniform spheres of mass M have radii R and $2R$. Each sphere is rotating about a fixed axis through a diameter. The rotational kinetic energies of the spheres are identical. What is the ratio of the magnitude of the angular momenta of these spheres? That is, $\frac{L_{2R}}{L_R} =$
- (A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$
29. A spinning ice skater can increase his rate of rotation by bringing his arms and free leg closer to his body. How does this procedure affect the skater's angular momentum and kinetic energy and what is the work done by the skater?
- (A) angular momentum remains the same while kinetic energy increases and work done is positive.
 (B) angular momentum remains the same while kinetic energy decreases and work done is negative.
 (C) both angular momentum and kinetic energy remain the same and work done is zero.
 (D) angular momentum increases while kinetic energy remains the same and work done may be positive or negative.
30. A child with mass m is standing at the edge of a disc with moment of inertia I , radius R , and initial angular velocity ω . See figure given below. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is



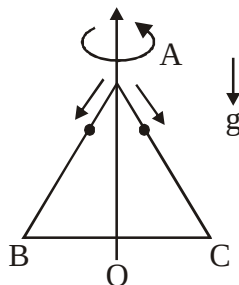
(A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$

(B) $\sqrt{\frac{(I + mR^2)\omega^2 - mv^2}{I}}$

(C) $\frac{I\omega - mvR}{I}$

(D) $\frac{(I + mR^2)\omega - mvR}{I}$

31. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are

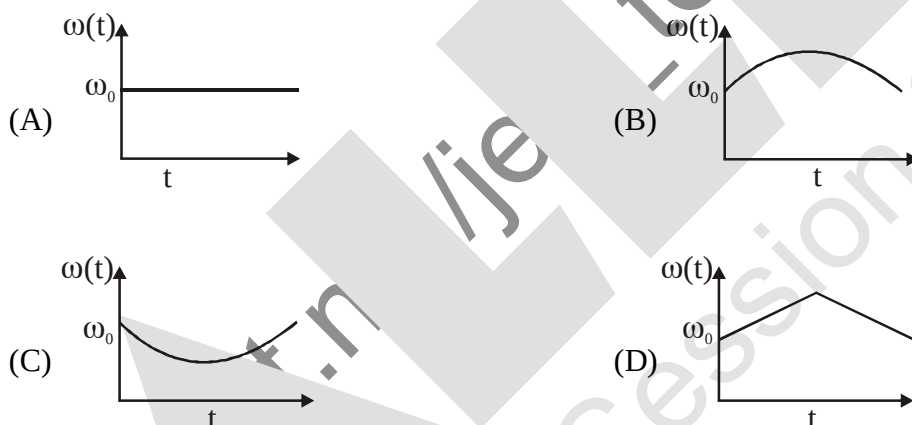


- (A) Angular velocity and total energy (kinetic and potential)
 (B) Total angular momentum and total energy
 (C) Angular velocity and moment of inertia about the axis of rotation.
 (D) Total angular momentum and moment of inertia about the axis of rotation.

[IIT-JEE 2000]

32. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0 . When the tortoise move along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$ will vary with time t as

[IIT-JEE 2002]

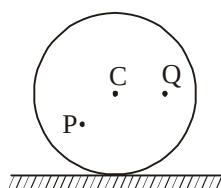


Combined rotation and translation

Kinematics

33. A disc is rolling without slipping with angular velocity ω . P and Q are two points equidistant from the centre C. The order of magnitude of velocity is :-

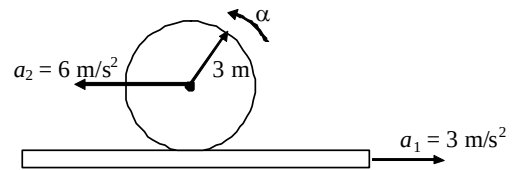
[IIT-JEE 2004]



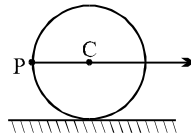
- (A) $v_Q > v_C > v_P$
 (B) $v_P > v_C > v_Q$
 (C) $v_P = v_C, v_Q = v_C/2$
 (D) $v_P < v_C > v_Q$

34. In the following figure, a sphere of radius 3 m rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is

- (A) 3 rad/s^2
 (B) 6 rad/s^2
 (C) 3 rad/s^2 (opposite to the direction shown in figure)
 (D) 1 rad/s^2



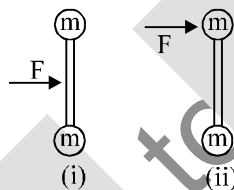
35. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is



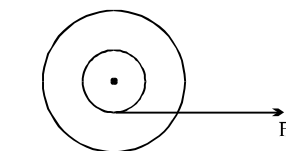
- (A) zero (B) 45° (C) 135° (D) $\tan^{-1}(1/2)$

Dynamics

36. A force F is applied to a dumbbell for a time interval, t , first as in (i) and then as in (ii). In which case does the dumbbell acquire the greater centre-of-mass speed?



- (A) (i)
 (B) (ii)
 (C) there is no difference
 (D) the answer depends on the rotational inertia of the dumbbell
37. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are
 (A) Up the incline while ascending and down the incline while descending [IIT-JEE 2002]
 (B) Up the incline while ascending as well as descending
 (C) down the incline while ascending and up the incline while descending
 (D) down the incline while ascending as well as descending
38. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in fig. then in case of pure rolling

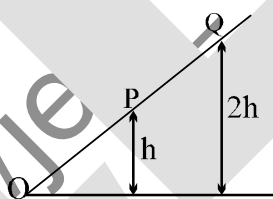


- (A) Thread unwinds, spool rotates anticlockwise and friction act leftwards
 (B) Thread winds, spool rotates clockwise and friction acts leftwards
 (C) Thread winds, spool moves to the right and friction act rightwards
 (D) Thread winds, spool moves to the right and friction does not come into existence.

39. A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :
 (A) thin pipe (B) uniform cylinder
 (C) uniform sphere (D) thin spherical shell
40. A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere:
 (A) must be forward (in direction of v) (B) must be backward (opposite to v)
 (C) cannot be zero (D) none of the above

Kinetic energy

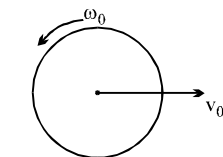
41. A hoop and a solid cylinder have the same mass and radius. They both roll, without slipping, on a horizontal surface. If their kinetic energies are equal
 (A) the hoop has a greater translational speed than the cylinder
 (B) the cylinder has a greater translational speed than the hoop
 (C) the hoop and the cylinder have the same translational speed
 (D) the hoop has a greater rotational speed than the cylinder
42. A ball rolls down an inclined plane as shown in figure. The ball is first released from rest from P and then later from Q. Which of the following statement is/ are correct?
 (i) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.
 (ii) The acceleration of the ball at Q is twice as large as the acceleration at P.
 (iii) The ball has twice as much K.E. at O when rolling from Q as it does when rolling from P.



- (A) i, ii only (B) ii, iii only (C) i only (D) iii only
43. The moment of inertia of a solid cylinder about its axis is given by $(1/2)MR^2$. If this cylinder rolls without slipping, the ratio of its rotational kinetic energy to its translational kinetic energy is
 (A) 1 : 1 (B) 2 : 2 (C) 1 : 2 (D) 1 : 3

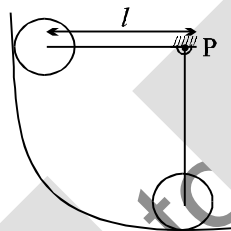
Angular momentum

44. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



- (A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

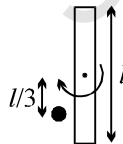
45. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m . It is rolled down a horizontal plane such that its centre of mass moves with a velocity v . If it purely rolls
- (A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$
- (B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$
- (C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}mvR$
- (D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}mvR$
46. A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P . The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is :



- (A) $M\sqrt{\frac{10}{7}gl} [l + R]$
- (B) $M\sqrt{\frac{10}{7}gl} \left[l + \frac{2}{5}R \right]$
- (C) $M\sqrt{\frac{10}{7}gl} \left[l + \frac{7}{5}R \right]$
- (D) $M\sqrt{\frac{10}{7}gl} \left[l - \frac{2}{5}R \right]$

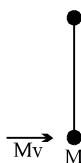
Eccentric collision

47. A uniform rod of length l and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is



- (A) $3M$
- (B) $\frac{3M}{4}$
- (C) $\frac{3M}{2}$
- (D) $\frac{4M}{3}$
48. A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is
- (A) $2v/d$
- (B) $2v/(3d)$
- (C) v/d
- (D) $3v/(2d)$

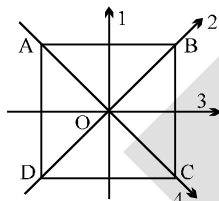
49. Two particles each of mass M are connected by a massless rod of length l . The rod is lying on the smooth surface. If one of the particle is given an impulse Mv as shown in the figure then angular velocity of the rod would be [IIT-JEE'(Scr)2003]



- (A) v/l (B) $2v/l$ (C) $v/2l$ (D) None

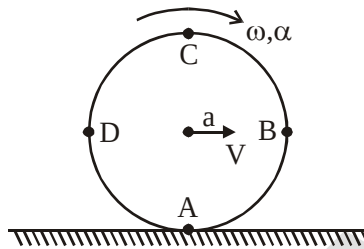
MULTIPLE CORRECT TYPE QUESTIONS

50. ABCD is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 & 4 are I_1 , I_2 , I_3 & I_4 respectively. It follows that :



- (A) $I_2 = I_3$ (B) $I = I_1 + I_4$ (C) $I = I_2 + I_4$ (D) $I_1 = I_3$
51. A rod of weight w is supported by two parallel knives at end points A and B and is in equilibrium in a horizontal position. The knives are at a distance ' d ' from each other. The centre of mass of the rod is at a distance ' x ' from A.
- (A) the normal reaction at A is $\frac{wx}{d}$ (B) the normal reaction at A is $\frac{w(d-x)}{d}$
- (C) the normal reaction at B is $\frac{wx}{d}$ (D) the normal reaction at B is $\frac{w(d-x)}{d}$
52. A block with a square base measuring ' $a \times a$ ' and height h , is placed on an inclined plane. The coefficient of friction is μ . The angle of inclination (θ) of the plane is gradually increased. The block will
- (A) topple before sliding if $\mu > \frac{a}{h}$ (B) topple before sliding if $\mu < \frac{a}{h}$
- (C) slide before toppling if $\mu > \frac{a}{h}$ (D) slide before toppling if $\mu < \frac{a}{h}$
53. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.
- (A) Angular momentum of the particle about O is increasing .
- (B) Torque of the gravitational force on the particle about O is decreasing.
- (C) The moment of inertia of the particle about O is decreasing .
- (D) The angular velocity of the particle about O is increasing.

54. A man spinning in free space changes the shape of his body, eg. by spreading his arms or curling up. By doing this, he can change his
 (A) moment of inertia (B) angular momentum
 (C) angular velocity (D) rotational kinetic energy
55. A circular disc of radius R rolls without slipping on a rough horizontal surface. At the instant shown its linear velocity is V , linear acceleration a , angular velocity ω and angular acceleration α . Four points A, B, C and D lie on its circumference such that the diameter AC is vertical & BD horizontal then choose the **CORRECT** option(s).



(A) $V_B = \sqrt{V^2 + (R\omega)^2}$

(B) $V_C = V + R\omega$

(C) $a_A = \sqrt{(a - R\alpha)^2 + (\omega^2 R)^2}$

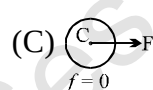
(D) $a_D = \sqrt{(a + \omega^2 R)^2 + (R\alpha)^2}$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 56 to 58

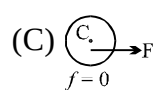
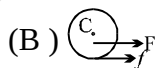
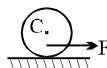
In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F .

56. A cylinder of mass M and radius R is pulled horizontally by a force F . The friction force can be given by which of the following diagrams :-



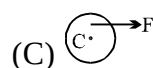
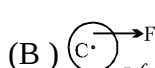
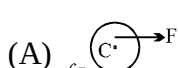
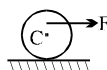
(D) cannot be interpreted

57. A cylinder is pulled horizontally by a force F acting at a point below the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams :-



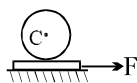
(D) cannot be interpreted

58. A cylinder is pulled horizontally by a force F acting at a point above the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



(D) cannot be interpreted

59. A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F . The friction force can be given by which of the following diagrams



- (A)  (B)  (C)  (D) cannot be interpreted

Paragraph for Question No. 60 to 62

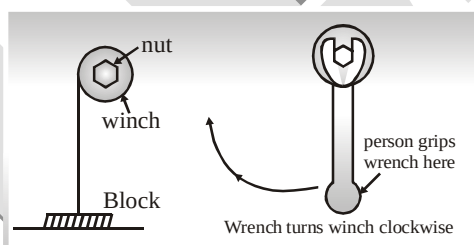
In figure, the winch is mounted on an axle, and the 6-sided nut is welded to the winch. By turning the nut with a wrench, a person can rotate the winch. For instance, turning the nut clockwise lifts the block off the ground, because more and more rope gets wrapped around the winch.

Three students agree that using a longer wrench makes it easier to turn the winch. But they disagree about why. All three students are talking about the case where the winch is used, over a 10 s time interval, to lift the block one metre off the ground.

Student 1 : By using a longer wrench, the person decreases the average force he must exert on the wrench, in order to lift the block one metre in 10 s.

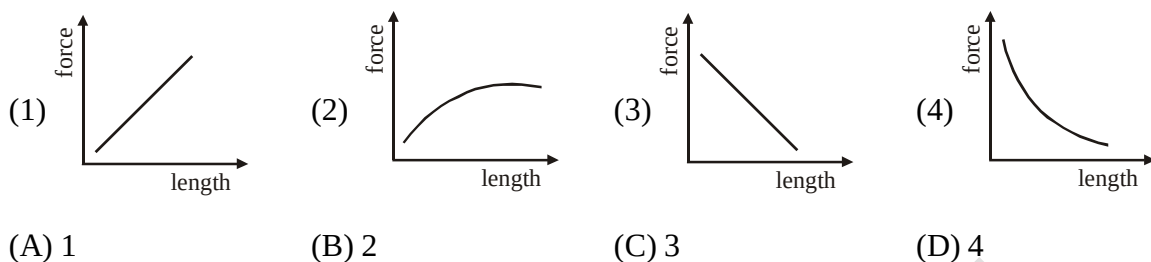
Student 2 : Using a longer wrench reduces the work done by the person as he uses the winch to lift the block 1m in 10s.

Student 3: Using a longer wrench reduces the power that the person must exert to lift the block 1m in 10s.



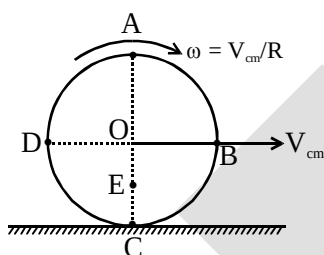
60. Student 1 is :-
- (A) correct, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
 - (B) correct, because using a longer wrench decreases the torque it must exert on the winch
 - (C) incorrect, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
 - (D) Incorrect, because using a longer wrench decreases the torque it must exert on the winch.
61. Which of the following is true about student 2 and 3 :-
- (A) Student 2 and 3 are both correct
 - (B) Student 2 is correct, but student 3 is incorrect
 - (C) Student 3 is correct, but student 2 is incorrect
 - (D) Student 2 and 3 are both incorrect

62. If several wrenches all apply the same torque to a nut, which graph best expresses the relationship between the force the person must apply to the wrench, and the length of the wrench :-



MATRIX MATCH TYPE QUESTION

63. Consider a body rolling on a horizontal surface as shown in figure (Symbols have their usual meaning)



Column-I

- (A) Velocity is zero
(B) Speed is maximum
(C) $0 < \text{speed} < v_{\text{cm}}$
(D) $1.3v_{\text{cm}} < \text{speed} < 2v_{\text{cm}}$

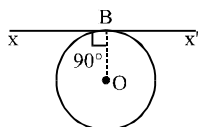
Column-II

- (P) Point A
(Q) Point B
(R) Point C
(S) Point D
(T) Point E

EXERCISE (O-2)

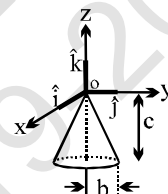
SINGLE CORRECT TYPE QUESTIONS

1. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is : **[IIT-JEE'(Scr)'2000]**



- (A) $\rho L^3/8\pi^2$ (B) $\rho L^3/16\pi^2$ (C) $5\rho L^3/16\pi^2$ (D) $3\rho L^3/8\pi^2$

2. A solid cone hangs from a frictionless pivot at the origin O , as shown. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors, and a , b , and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_z ?



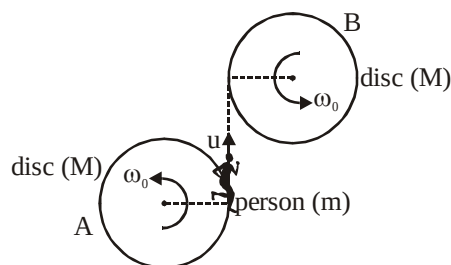
- (A) $F = a\hat{k}$, P is $(0, b, -c)$ (B) $F = -a\hat{k}$, P is $(0, -b, -c)$
(C) $F = a\hat{j}$, P is $(-b, 0, -c)$ (D) None

3. A heavy seesaw (i.e., not massless) is out of balance. A light girl sits on the end that is tilted downward, and a heavy body sits on the other side so that the seesaw now balances. If they both move forward so that they are one-half their original distance from the pivot point (the fulcrum) what will happen to the seesaw ?

- (A) The side the body is sitting on will tilt downward
(B) The side the girl is sitting on will once again tilt downward
(C) Nothing ; the seesaw will still be balanced
(D) It is impossible to say without knowing the masses and the distances

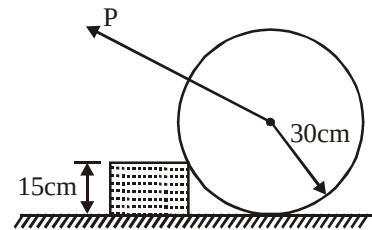
4. Two identical discs each of mass $M (= 4m)$ and radius R are rotating in opposite sense with equal angular speed ω_0 about vertical axes passing through their centres, (as shown in figure). A person of mass m sitting on circumference of disc A jumps with a tangential relative velocity u (after jumping) w.r.t one rotating disc (A) and lands on other disc (B) also tangential. Now the second disc (B) comes to a stop. Find the relative velocity u .

- (A) $u = \frac{\omega_0 R}{2}$
(B) $u = \omega_0 R$
(C) $u = \frac{3\omega_0 R}{2}$
(D) $u = 2\omega_0 R$



5. In the given figure a uniform wheel of radius 30cm rests against a rigid rectangular block 15cm high. The wheel weighs 1000 N. The minimum pull P through the center which will turn the wheel over the block is :-

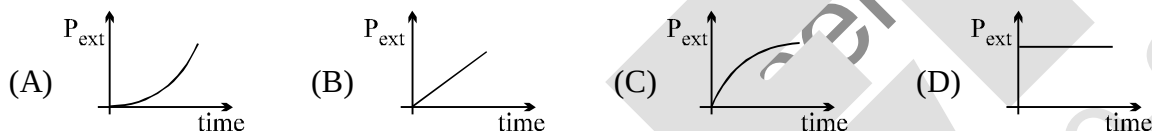
(A) $500\sqrt{3}$ N (B) $1000\sqrt{3}$ N
(C) 1000 N (D) $400\sqrt{3}$ N



6. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of particle P at a distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is

(A) $2\frac{J}{m}$ (B) $\frac{J}{\sqrt{2}m}$ (C) $\frac{J}{m}$ (D) $\sqrt{2}\frac{J}{m}$

7. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :



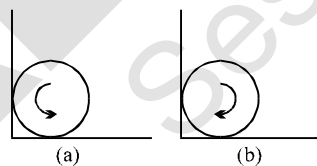
8. A straight rod of length L is released on a frictionless horizontal floor in a vertical position. As it falls + slips, the distance of a point on the rod from the lower end, which follows a quarter circular locus is :-

(A) $L/2$ (B) $L/4$ (C) $L/8$ (D) None

9. A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) & (b).

Coefficient of friction between all surfaces and the sphere is $1/3$. Find the ratio of the frictional force $\frac{f_a}{f_b}$

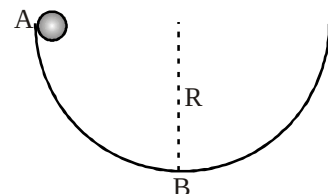
by ground in situations (a) & (b).



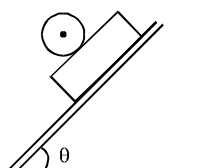
(A) 1 (B) $\frac{9}{10}$ (C) $\frac{10}{9}$ (D) none

10. A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R ($\gg r$) as shown in figure. If the sphere starts from rest at the top point of the hemisphere find the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere.

(A) $\frac{10}{7}mg$ (B) $\frac{17}{7}mg$ (C) $\frac{5}{7}mg$ (D) $\frac{7}{5}mg$

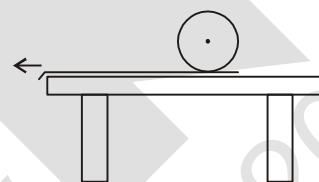


- 11.** A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is:

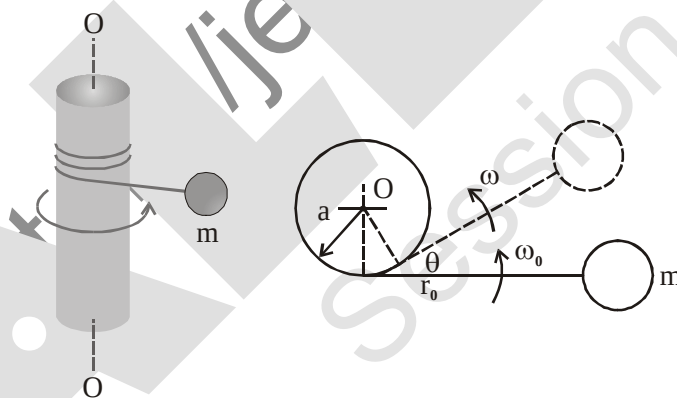


- (A) up the plane (B) down the plane (C) horizontal (D) zero
- 12.** A thin table cloth covers a horizontal table and a uniform body of round shape lies on top of it. The table cloth is pulled from under the body, and friction causes the body to slide and rotate. What is the body's final motion on the table? (Assume that the table is so large that the body does not fall off it.)

- (A) Body will finally roll towards left
(B) Body will finally roll towards right
(C) Body will finally come to rest
(D) Any of the above is possible depending on shape of body.



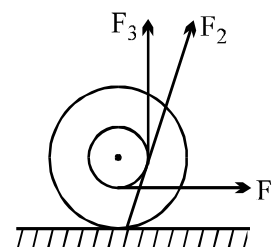
13. The small particle of mass m is given an initial high velocity in the horizontal plane and winds its cord around the fixed vertical shaft of radius 1 m . All motion occurs essentially in the horizontal plane. If the angular velocity of the cord is 0.8 rad/s when the distance from the particle to the tangential point is 5 m , determine the angular velocity ω (in rad/s) of the cord after it has turned through an angle 1 rad .



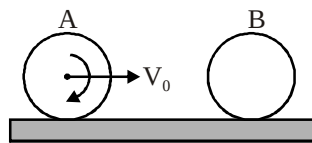
- (A) 1 (B) 2 (C) 4 (D) 6

14. A yo-yo is resting on a perfectly rough horizontal table. Forces F_1 , F_2 and F_3 are applied separately as shown. The correct statement is :-

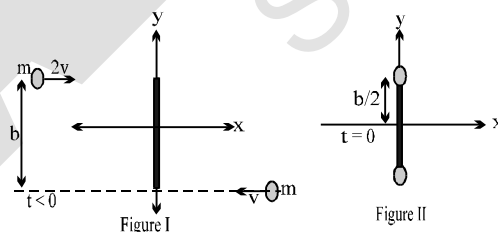
- (A) when F_3 is applied the centre of mass will move to the right.
(B) when F_2 is applied the centre of mass will move to the left.
(C) when F_1 is applied the centre of mass will move to the right.
(D) when F_2 is applied the centre of mass will move to the right.



15. A hollow smooth uniform sphere A of mass 'm' rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of 'B' to that of 'A' just after the collision is :



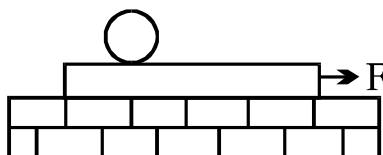
- (A) 1 : 1 (B) 2 : 3 (C) 3 : 2 (D) None
16. A wheel of radius r rolling on a straight line, the velocity of its centre being v . At a certain instant the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The incorrect statement is:
- (A) The velocity of any point P of the wheel is proportional to MP.
 (B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v .
 (C) The point of contact M is instantaneously at rest.
 (D) The velocities of any two parts of the wheel which are equidistant from centre are equal.
17. A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30^\circ$ with the horizontal. Then the speed of the ladder's center must be :-
- (A) $2v/\sqrt{3}$ (B) $v/2$ (C) v (D) None
18. In the previous question, if $dv/dt = 0$, then the angular acceleration of the ladder when $\alpha = 45^\circ$ is :-
- (A) $2v^2/L^2$ (B) $v^2/2L^2$ (C) $\sqrt{2}[v^2/L^2]$ (D) None
19. One ice skater of mass m moves with speed $2v$ to the right, while another of the same mass m moves with speed v toward the left, as shown in figure I. Their paths are separated by a distance b . At $t = 0$, when they are both at $x = 0$, they grasp a pole of length b and negligible mass. For $t > 0$, consider the system as a rigid body of two masses m separated by distance b , as shown in figure II. Which of the following is the correct formula for the motion after $t = 0$ of the skater initially at $y = b/2$?



- (A) $x = 2vt, y = b/2$
 (B) $x = vt + 0.5b \sin(3vt/b), y = 0.5b \cos(3vt/b)$
 (C) $x = 0.5vt + 0.5b \sin(3vt/b), y = 0.5b \cos(3vt/b)$
 (D) $x = 0.5vt + 0.5b \sin(6vt/b), y = 0.5b \cos(6vt/b)$

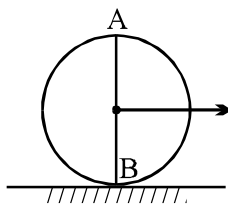
MULTIPLE CORRECT TYPE QUESTIONS

20. A rigid object is rotating in a counterclockwise sense around a fixed axis. If the rigid object rotates through more than 180° but less than 360° , which of the following pairs of quantities can represent an initial angular position and a final angular position of the rigid object. Which of the sets can only occur.
- (A) 3 rad, 6 rad (B) -1 rad, 1 rad (C) 1 rad, 5 rad (D) -1 rad, 2.5 rad
21. A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is
- (A) 2, if their lines of action pass through the centre of mass of the body.
 (B) 3, if their lines of action are not parallel.
 (C) 3, if their lines of action are parallel.
 (D) 4, if their lines of action are parallel and all the forces have the same magnitude.
22. The torque $\vec{\tau}$ on a body about a given point is found to be equal to $\vec{A} \times \vec{L}$ where $\vec{A}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this it follows that
- (A) $d\vec{L}/dt$ is perpendicular to $\vec{L}$ at all instants of time
 (B) the components of $\vec{L}$ in the direction of $\vec{A}$ does not change with time
 (C) the magnitude of $\vec{L}$ does not change with time
 (D) $\vec{L}$ does not change with time
23. A block of mass m moves on a horizontal rough surface with initial velocity v . The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.
- (A) angular momentum about A is mvh initially
 (B) the velocity of the block decreases at time passes.
 (C) torque of the forces acting on block is zero about A
 (D) angular momentum is not conserved about A.
24. If a cylinder is rolling down the incline with sliding.
- (A) after some time it may start pure rolling
 (B) after sometime it will start pure rolling
 (C) it may be possible that it will never start pure rolling
 (D) none of these
25. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F . If sphere does not slip over the plank. Which of the following is correct.

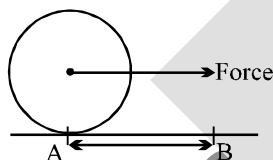


- (A) Acceleration of the centre of sphere is less than that of the plank.
 (B) Work done by friction acting on the sphere is equal to its total kinetic energy.
 (C) Total kinetic energy of the system is equal to work done by the force F
 (D) None of the above

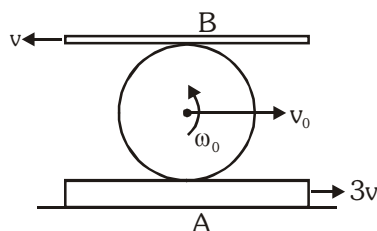
26. A uniform disc is rolling on a horizontal surface. At a certain instant B is the point of contact and A is at height $2R$ from ground, where R is radius of disc.



- (A) The magnitude of the angular momentum of the disc about B is thrice that about A.
 (B) The angular momentum of the disc about A is anticlockwise.
 (C) The angular momentum of the disc about B is clockwise
 (D) The angular momentum of the disc about A is equal to that about B.
27. A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. $AB = s$. The disc moves from A to B in time T . To the right of B,

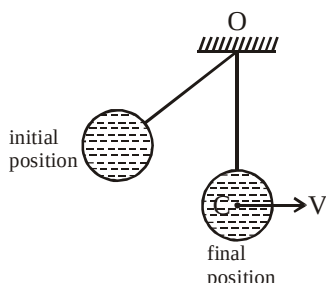


- (A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged
 (B) linear acceleration of the disc will increase
 (C) the disc will make one rotation in time $T/2$
 (D) the disc will cover a distance greater than s in further time T .
28. A ring rolls without slipping on the ground. Its centre C moves with a constant speed u . P is any point on the ring. The speed of P with respect to the ground is v .
 (A) $0 \leq v \leq 2u$
 (B) $v = u$, if CP is horizontal
 (C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
 (D) $v = \sqrt{2}u$, if CP is horizontal
29. The disc of radius r is confined to roll without slipping at A and B. If the plates have the velocities shown, then

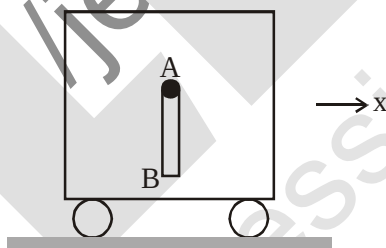


- (A) linear velocity $v_0 = v$ (B) angular velocity of disc is $\frac{3v}{2r}$
 (C) angular velocity of disc is $\frac{2v}{r}$ (D) None of these

30. A massless rod has a massless hollow sphere attached to it. This sphere can be fully filled either with a liquid (non viscous) or with a solid (rigidly fitted into sphere) of same mass. System is released from rest from initial position (as shown). When it reaches final position which of the following is/are true for the system.



- (A) Kinetic energy in case of liquid will be more than in case of solid.
 (B) Velocity of centre (V) in case of liquid will be more than in case of solid.
 (C) About C angular momentum in case of liquid will be more than in case of solid.
 (D) About C angular momentum in case of liquid will be less than in case of solid.
31. A uniform rod AB of length ℓ and mass M hangs from point A at which it is freely hinged in a car moving with velocity v_0 . The rod can rotate in vertical plane about the axis at A. If the car suddenly stops,

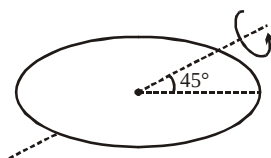


- (A) The angular speed ω with which the rod starts rotating is $\frac{3v_0}{2\ell}$
 (B) The minimum value of v_0 so that the rod completes the rotation $\sqrt{\frac{8}{3}g\ell}$
 (C) Loss of energy during the process $\frac{1}{8}Mv_0^2$
 (D) There is no loss of energy

36. Consider a solid cube of mass m and side L . What will be the value of $\sum m_i r_i^2$ for this body when the point of intersection of axes is the centre of the cube :

(A) $\frac{ML^2}{2}$ (B) $\frac{ML^2}{4}$ (C) $\frac{ML^2}{3}$ (D) $\frac{ML^2}{6}$

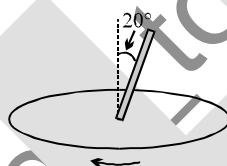
37. Find the moment of inertia of ring of mass m and radius R about an axis passing through its centre and making an angle 45° with its plane :



(A) $\frac{MR^2}{4}$ (B) $\frac{MR^2}{2}$ (C) $\frac{3}{4}MR^2$ (D) MR^2

Paragraph for Question No. 38 and 39

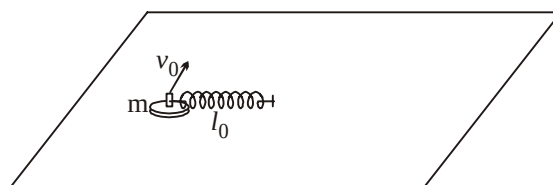
A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) If the turntable is rotating clockwise as seen from above.



38. What is the direction of the rod's angular momentum vector (calculated about its lower end)?
 (A) vertically downwards (B) down at 20° to the horizontal
 (C) up at 20° to the horizontal (D) vertically upwards
39. Is there a torque acting on it, and if so in what direction?
 (A) yes, vertically (B) yes, horizontally
 (C) yes at 20° to the horizontal (D) no

Paragraph for Question No. 40 to 43

A spring having initial unstretched length ℓ_0 is lying on a smooth table. It's one end is fixed and the other one is fastened to a small particle of mass m . The particle is imparted an initial speed v_0 horizontally in a direction perpendicular to the spring. In the course of the motion in horizontal plane, the maximum elongation of the spring is $\Delta\ell = \ell_0/10$. (Given : $m = 0.1$ kg, $\ell_0 = 1$ m, $v_0 = 11$ m/s).



40. In the course of motion, which of the quantities relating to spring block system are conserved ?
 (A) kinetic energy (B) momentum (C) angular momentum (D) potential energy

41. Which of the following is correct about initial situation and situation at maximum elongation ?
 (A) the orientation of spring in both positions should be perpendicular to each other.
 (B) the velocity at maximum extension should be zero
 (C) the velocity at maximum extension as well as at initial position should be perpendicular to spring.
 (D) Acceleration should be zero at the maximum extension as well as at initial position.
42. **Student-A** : at maximum extension $mv'(\ell_0 + \Delta\ell) = mv_0\ell_0$.

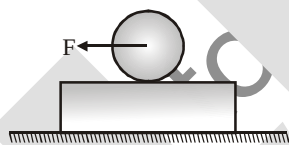
Student-B : at maximum extension $k\Delta\ell = \frac{mv'^2}{(\ell_0 + \Delta\ell)}$. **Student-C** : $\frac{1}{2}mv_0^2 = \frac{1}{2}mv'^2 + \frac{1}{2}k\Delta\ell^2$.

where v' is velocity at instant of maximum extension :

- (A) Only Student-A and B are correct (B) Only Student-A and C are correct
 (C) Only Student-B and C are correct (D) All are correct
43. The value of spring constant (in N/m) is :
 (A) 100 (B) 210 (C) 420 (D) 105

Paragraph for Question No. 44 and 45

A disc of mass m and radius R is placed over a plank of same mass m . There is sufficient friction between disc and plank to prevent slipping. A force F is applied at the centre of the disc.

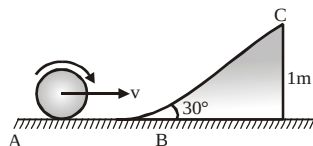


44. Acceleration of the plank is :-
 (A) $\frac{F}{2m}$ (B) $\frac{3F}{4m}$ (C) $\frac{F}{4m}$ (D) $\frac{3F}{2m}$
45. Force of friction between the disc and the plank is :-
 (A) $\frac{F}{2}$ (B) $\frac{F}{4}$ (C) $\frac{F}{3}$ (D) $\frac{2F}{3}$

Paragraph for Question No. 46 and 47

A small sphere of mass 1 kg is rolling without slipping on a rough stationary base with linear speed

$v = \sqrt{\frac{200}{7}}$ m/s. It leaves the inclined plane at point C.



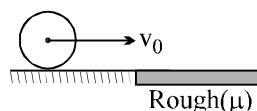
46. Find its linear speed at point C :-
 (A) $\sqrt{\frac{100}{7}}$ m/s (B) $\sqrt{\frac{50}{7}}$ m/s (C) $\sqrt{\frac{100}{35}}$ m/s (D) $\sqrt{\frac{200}{35}}$ m/s

47. Find ratio of rotational and translational kinetic energy of the sphere when it strikes the ground after leaving from point C :-

- (A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{2}$

Paragraph for Question No. 48 to 51

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ , as shown in figure.



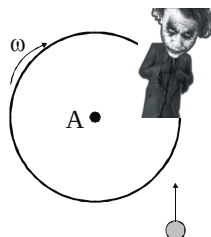
48. Choose the correct statement(s)
- (A) As the ring enters on the rough surface, the limiting friction force acts on it
- (B) The direction of friction is opposite to the direction of motion
- (C) The friction force accelerates the ring in the clockwise sense about its centre of mass
- (D) As the ring enters on the rough surface it starts rolling
49. Choose the correct statement(s)
- (A) The momentum of the ring is conserved
- (B) The angular momentum of the ring is conserved about its centre of mass
- (C) The angular momentum of the ring conserved about any point on the horizontal surface in line of friction.
- (D) The mechanical energy of the ring is conserved
50. Choose the correct statement(s)
- (A) The ring starts its rolling motion when the centre of mass stationary
- (B) The ring starts rolling motion when the point of contact becomes stationary
- (C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$
- (D) The rolling velocity is $\frac{v_0}{2}$
51. Choose the correct alternative(s) :-
- (A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$
- (B) The net work done by friction force is $-\frac{3}{8}mv_0^2$
- (C) The loss in kinetic energy of the ring is $\frac{mv_0^2}{4}$
- (D) The gain in rotational kinetic energy is $+\frac{mv_0^2}{8}$

MATRIX MATCH TYPE QUESTION

52. Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I.

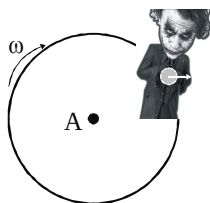
Column-I

- (A) Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.



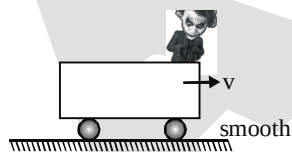
Joker, ball and platform is system.

- (B) Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.



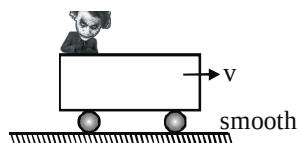
Joker, ball and platform is system.

- (C) Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.



Joker and cart is the system

- (D) Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.



Joker and cart is the system

Column-II

- (P) Linear momentum remains conserved.

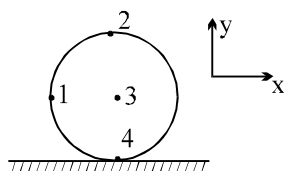
- (Q) Mechanical energy is conserved

- (R) Mechanical energy increases.

- (S) Mechanical energy decreases.

- (T) v or ω changes

53. A rigid cylinder is kept on a smooth horizontal surface as shown. If **Column-I** indicates velocities of various points (3-centre of cylinder, 2- top point, 4-bottom point, 1- on the level of 3 at the rim) on it shown, choose correct state of motion from **Column-II**.



Column-I

- (A) $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_2 = 2\hat{i}$
 (B) $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_3 = -\hat{i}$
 (C) $\vec{v}_2 = \hat{i}, \vec{v}_3 = 0$
 (D) $\vec{v}_4 = 0, \vec{v}_1 = -\hat{i} - \hat{j}$

Column-II

- (P) Pure rotation about centre
 (Q) Rolling without slipping to left
 (R) Rolling without slipping to right
 (S) Not possible

EXERCISE (JM)

1. A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω . Its centre of mass rises to a maximum height of:-

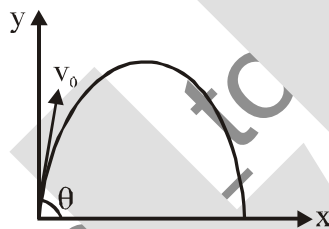
[AIEEE - 2009]

(1) $\frac{1}{2} \frac{l^2 \omega^2}{g}$ (2) $\frac{1}{6} \frac{l^2 \omega^2}{g}$ (3) $\frac{1}{3} \frac{l^2 \omega^2}{g}$ (4) $\frac{1}{6} \frac{l \omega}{g}$

2. A small particle of mass m is projected at an angle θ with the x -axis with an initial velocity v_0 in the x - y plane as shown in the figure. At a time $t < \frac{v_0 \sin \theta}{g}$, the angular momentum of the particle is:

Where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x , y and z -axis respectively.

[AIEEE-2010]



(1) $\frac{1}{2} mg v_0 t^2 \cos \theta \hat{i}$ (2) $-mg v_0 t^2 \cos \theta \hat{j}$
 (3) $mg v_0 t \cos \theta \hat{k}$ (4) $-\frac{1}{2} mg v_0 t^2 \cos \theta \hat{k}$

3. A pulley of radius 2 m is rotated about its axis by a force $F = (20t - 5t^2)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is 10 kg m^2 , the number of rotations made by the pulley before its direction of motion it reversed, is :-

(1) more than 6 but less than 9 (2) more than 9 [AIEEE-2011]
 (3) less than 3 (4) more than 3 but less than 6

4. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, then angular speed of the disc :-

[AIEEE-2011]

(1) continuously increases (2) first increases and then decreases
 (3) remains unchanged (4) continuously decreases

5. A particle of mass 'm' is projected with a velocity v making an angle of 30° with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height 'h' is :- [AIEEE-2011]

(1) $\frac{\sqrt{3}}{2} \frac{mv^2}{g}$ (2) zero (3) $\frac{mv^3}{\sqrt{2}g}$ (4) $\frac{\sqrt{3}}{16} \frac{mv^3}{g}$

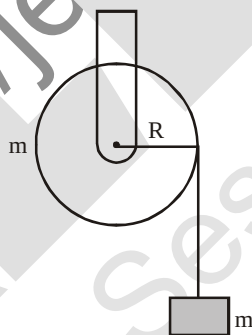
6. A hoop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip? [JEE Mains-2013]

(1) $\frac{r\omega_0}{4}$ (2) $\frac{r\omega_0}{3}$ (3) $\frac{r\omega_0}{2}$ (4) $r\omega_0$

7. A bob of mass m attached to an inextensible string of length ℓ is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed ω rad/s about the vertical. About the point of suspension : [JEE Mains-2014]

- (1) Angular momentum changes in direction but not in magnitude
(2) Angular momentum changes both in direction and magnitude
(3) Angular momentum is conserved
(4) Angular momentum changes in magnitude but not in direction.

8. A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R . If the string does not slip on the cylinder, with what acceleration will the mass fall on release? [JEE Mains-2014]

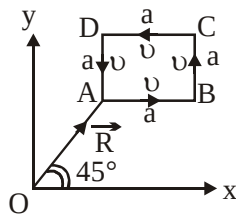


(1) $\frac{5g}{6}$ (2) g (3) $\frac{2g}{3}$ (4) $\frac{g}{2}$

9. From a solid sphere of mass M and radius R a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its centre and perpendicular to one of its faces is:- [JEE Mains-2015]

(1) $\frac{4MR^2}{9\sqrt{3}\pi}$ (2) $\frac{4MR^2}{3\sqrt{3}\pi}$ (3) $\frac{MR^2}{32\sqrt{2}\pi}$ (4) $\frac{MR^2}{16\sqrt{2}\pi}$

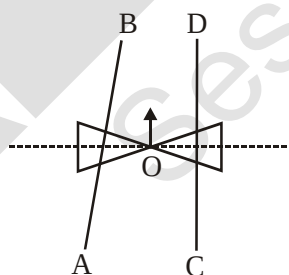
10. A particle of mass m is moving along the side of a square of side ' a ', with a uniform speed v in the x - y plane as shown in the figure : [JEE Mains-2016]



Which of the following statement is false for the angular momentum $\vec{L}$ about the origin ?

- (1) $\vec{L} = \frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from D to A
- (2) $\vec{L} = -\frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from A to B
- (3) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} - a \right] \hat{k}$ when the particle is moving from C to D
- (4) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} + a \right] \hat{k}$ when the particle is moving from B to C
11. A roller is made by joining together two cones at their vertices O. It is kept on two rails AB and CD which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to :-

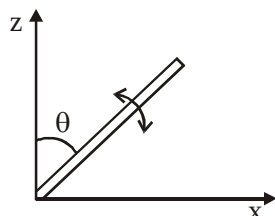
[JEE Mains-2016]



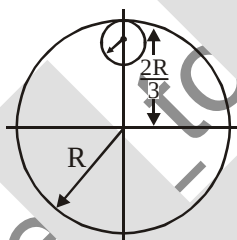
- (1) turn left and right alternately. (2) turn left.
- (3) turn right. (4) go straight.
12. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I . What is the ratio ℓ/R such that the moment of inertia is minimum ? [JEE Main-2017]

- (1) 1 (2) $\frac{3}{\sqrt{2}}$ (3) $\sqrt{\frac{3}{2}}$ (4) $\frac{\sqrt{3}}{2}$

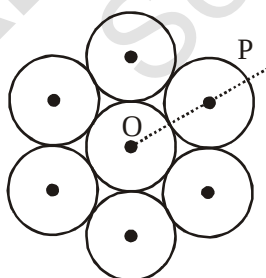
13. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is : [JEE Main-2017]



- (1) $\frac{3g}{2\ell} \cos \theta$ (2) $\frac{2g}{3\ell} \cos \theta$ (3) $\frac{3g}{2\ell} \sin \theta$ (4) $\frac{2g}{3\ell} \sin \theta$
14. From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is : [JEE Main-2018]



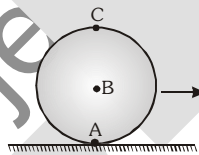
- (1) $\frac{40}{9} MR^2$ (2) $10 MR^2$ (3) $\frac{37}{9} MR^2$ (4) $4 MR^2$
15. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is : [JEE Main-2018]



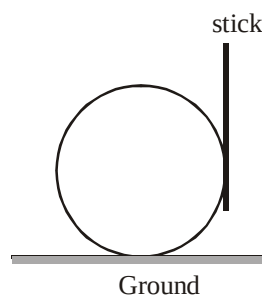
- (1) $\frac{55}{2} MR^2$ (2) $\frac{73}{2} MR^2$ (3) $\frac{181}{2} MR^2$ (4) $\frac{19}{2} MR^2$

EXERCISE (JA)

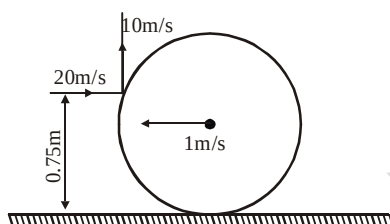
1. A block of base $10 \text{ cm} \times 10 \text{ cm}$ and height 15 cm is kept on an inclined plane. The coefficient of friction between them is $\sqrt{3}$. The inclination θ of this inclined plane from the horizontal plane is gradually increased from 0° . Then :- [IIT-JEE 2009]
 - (A) at $\theta = 30^\circ$, the block will start sliding down the plane
 - (B) the block will remain at rest on the plane up to certain θ and then it will topple
 - (C) at $\theta = 60^\circ$, the block will start sliding down the plane and continue to do so at higher angles
 - (D) at $\theta = 60^\circ$, the block will start sliding down the plane and on further increasing θ , it will topple at certain θ
2. If the resultant of the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that [IIT-JEE 2009]
 - (A) linear momentum of the system does not change in time
 - (B) kinetic energy of the system does not change in time
 - (C) angular momentum of the system does not change in time
 - (D) potential energy of the system does not change in time
3. A sphere is rolling without slipping on a fixed horizontal plane surface. In the figure, A is the point of contact, B is the centre of the sphere and C is its topmost point. Then [IIT-JEE 2009]



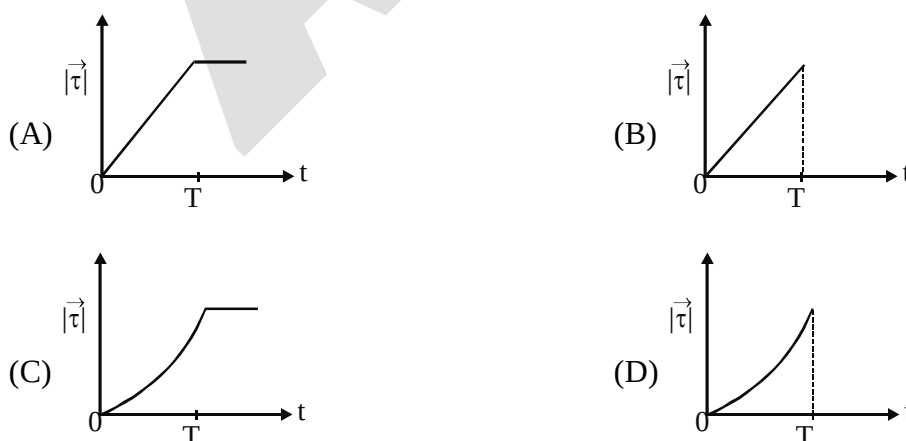
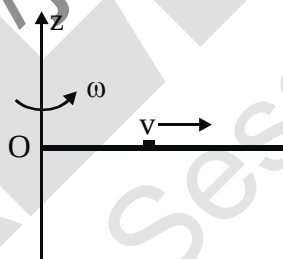
- (A) $\vec{v}_C - \vec{v}_A = 2(\vec{v}_B - \vec{v}_C)$
 - (B) $\vec{v}_C - \vec{v}_B = \vec{v}_B - \vec{v}_A$
 - (C) $|\vec{v}_C - \vec{v}_A| = 2|\vec{v}_B - \vec{v}_C|$
 - (D) $|\vec{v}_C - \vec{v}_A| = 4|\vec{v}_B|$
4. A boy is pushing a ring of mass 2 kg and radius 0.5 m with a stick as shown in the figure. The stick applies a force of 2 N on the ring and rolls it without slipping with an acceleration of 0.3 m/s^2 . The coefficient of friction between the ground and the ring is large enough that rolling always occurs and the coefficient of friction between the stick and the ring is $(P/10)$. The value of P is [IIT-JEE 2011]



5. Four solid spheres each of diameter $\sqrt{5}$ cm and mass 0.5 kg are placed with their centers at the corners of a square of side 4 cm. The moment of inertia of the system about the diagonal of the square is $N \times 10^{-4}$ kg-m², then N is [IIT-JEE 2011]
6. A thin ring of mass 2 kg and radius 0.5 m is rolling without slipping on a horizontal plane with velocity 1 m/s. A small ball of mass 0.1 kg, moving with velocity 20 m/s in the opposite direction, hits the ring at a height of 0.75 m and goes vertically up with velocity 10 m/s. Immediately after the collision [IIT-JEE 2011]

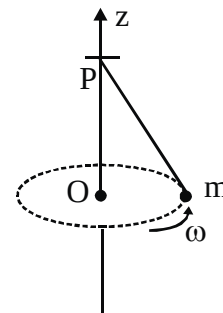


- (A) the ring has pure rotation about its stationary CM
 (B) the ring comes to a complete stop
 (C) friction between the ring and the ground is to the left
 (D) there is no friction between the ring and the ground
7. A thin uniform rod, pivoted at OP, is rotating in the horizontal plane with constant angular speed ω , as shown in the figure. At time $t = 0$, a small insect starts from O and moves with constant speed v with respect to the rod towards the other end. If it reaches the end of the rod at $t = T$ and stops. The angular speed of the system remains ω throughout. The magnitude of the torque ($|\tau|$) on the system about O, as a function of time is best represented by which plot? [IIT-JEE 2012]



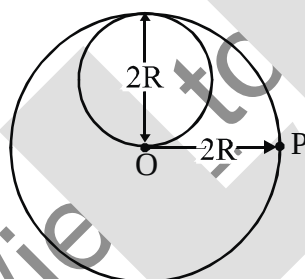
8. A small mass m is attached to a massless string whose other end is fixed at P as shown in the figure. The mass is undergoing circular motion in the x - y plane with centre at O and constant angular speed ω . If the angular momentum of the system, calculated about O and P are denoted by $\vec{L}_O$ and $\vec{L}_P$ respectively, then

[IIT-JEE 2012]



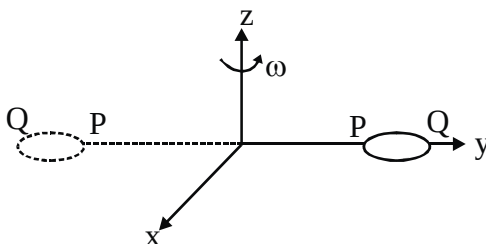
- (A) $\vec{L}_O$ and $\vec{L}_P$ do not vary with time
 (B) $\vec{L}_O$ varies with time while $\vec{L}_P$ remains constant
 (C) $\vec{L}_O$ remains constant while $\vec{L}_P$ varies with time
 (D) $|\vec{L}_O|$ and $|\vec{L}_P|$ both do not vary with time
9. A lamina is made by removing a small disc of diameter $2R$ from a bigger disc of uniform mass density and radius $2R$, as shown in the figure. The moment of inertia of this lamina about axes passing through O and P is I_O and I_P respectively. Both these axes are perpendicular to the plane of the lamina. The ratio $\frac{I_P}{I_O}$ to the nearest integer is

[IIT-JEE 2012]

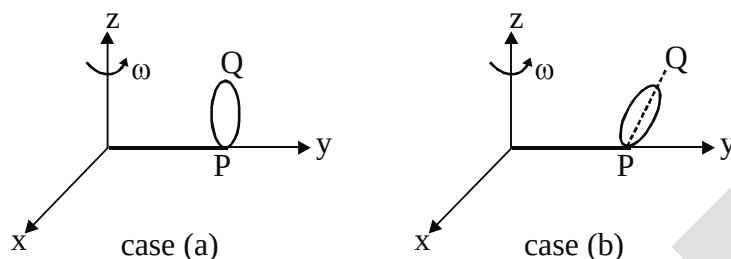


Paragraph for Questions 10 and 11

The general motion of a rigid body can be considered to be a combination of (i) a motion of its centre of mass about an axis, and (ii) its motion about an instantaneous axis passing through the centre of mass. These axes need not be stationary. Consider, for example, a thin uniform disc welded (rigidly fixed) horizontally at its rim to a massless stick, as shown in the figure. When the disc-stick system is rotated about the origin on a horizontal frictionless plane with angular speed ω , the motion at any instant can be taken as a combination of (i) a rotation of the centre of mass of the disc about the z -axis, and (ii) a rotation of the disc through an instantaneous vertical axis passing through its centre of mass (as is seen from the changed orientation of points P and Q). Both these motions have the same angular speed ω in this case.



Now consider two similar systems as shown in the figure : case (A) the disc with its face vertical and parallel to x-z plane; Case (B) the disc with its face making an angle of 45° with x-y plane and its horizontal diameter parallel to x-axis. In both the cases, the disc is welded at point P, and the systems are rotated with constant angular speed ω about the z-axis. [IIT-JEE 2012]



10. Which of the following statements regarding the angular speed about the instantaneous axis (passing through the centre of mass) is correct?

(A) It is $\sqrt{2\omega}$ for both the cases. (B) It is ω for case (a); and $\frac{\omega}{\sqrt{2}}$ for case (b).
(C) It is ω for case (a); and $\sqrt{2\omega}$ for case (b). (D) It is ω for both the cases.

11. Which of the following statements about the instantaneous axis (passing through the centre of mass) is correct?

(A) It is vertical for both the cases (a) and (b).
(B) It is vertical for case (a); and is at 45° to the x-z plane and lies in the plane of the disc for case (b).
(C) It is horizontal for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).
(D) It is vertical for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).

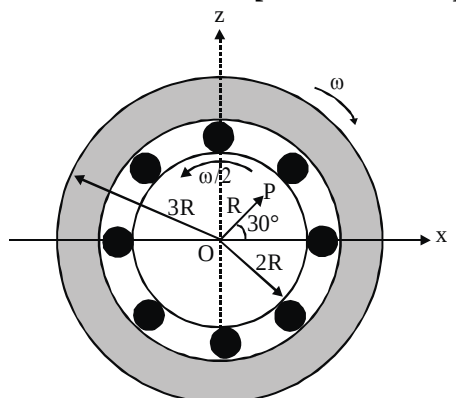
12. The figure shows a system consisting of (i) a ring of outer radius $3R$ rolling clockwise without slipping on a horizontal surface with angular speed ω and (ii) an inner disc of radius $2R$ rotating anti-clockwise with angular speed $\omega/2$. The ring and disc are separated by frictionless ball bearing. The system is in the x-z plane. The point P on the inner disc is at a distance R from the origin, where OP makes an angle of 30° with the horizontal. Then with respect to the horizontal surface, [IIT-JEE 2012]

(A) the point O has a linear velocity $3R\omega\hat{i}$

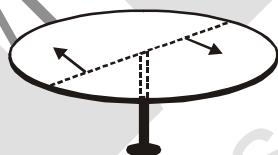
(B) the point P has a linear velocity $\frac{11}{4}R\omega\hat{i} + \frac{\sqrt{3}}{4}R\omega\hat{k}$

(C) the point P has a linear velocity $\frac{13}{4}R\omega\hat{i} - \frac{\sqrt{3}}{4}R\omega\hat{k}$

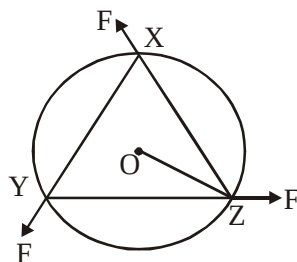
(D) the point P has a linear velocity $\left(3 - \frac{\sqrt{3}}{4}\right)R\omega\hat{i} + \frac{1}{4}R\omega\hat{k}$



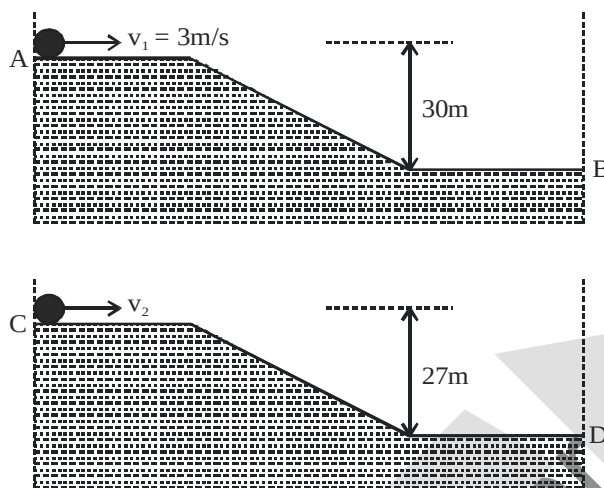
13. Two solid cylinders P and Q of same mass and same radius start rolling down a fixed inclined plane from the same height at the same time. Cylinder P has most of its mass concentrated near its surface, while Q has most of its mass concentrated near the axis. Which statement(s) is(are) correct?
 (A) Both cylinders P and Q reach the ground at the same time. [IIT-JEE 2012]
 (B) Cylinder P has larger acceleration than cylinder Q.
 (C) Both cylinders reach the ground with same translational kinetic energy.
 (D) Cylinder Q reaches the ground with larger angular speed.
14. A uniform circular disc of mass 50 kg and radius 0.4 m is rotating with an angular velocity of 10 rad s^{-1} about its own axis, which is vertical. Two uniform circular rings, each of mass 6.25 kg and radius 0.2 m, are gently placed symmetrically on the disc in such a manner that they are touching each other along the axis of the disc and are horizontal. Assume that the friction is large enough such that the rings are at rest relative to the disc and the system rotates about the original axis. The new angular velocity (in rad s^{-1}) of the system is [IIT-JEE 2013]
15. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is [JEE Advanced-2014]



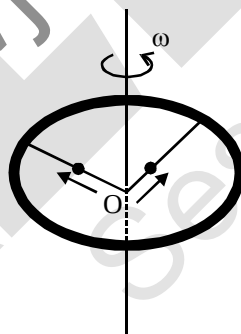
16. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5 \text{ N}$ are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is [JEE Advanced-2014]



17. Two identical uniform discs roll without slipping on two different, surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2 respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and $v_1 = 3\text{ m/s}$, then v_2 in m/s is ($g = 10\text{ m/s}^2$) **[JEE Advanced-2015]**



18. A ring of mass M and radius R is rotating with angular speed ω about a fixed vertical axis passing through its centre O with two point masses each of mass $\frac{M}{8}$ at rest at O . These masses can move radially outwards along two massless rods fixed on the ring as shown in the figure. At some instant the angular speed of the system is $\frac{8}{9}\omega$ and one of the masses is at a distance of $\frac{3}{5}R$ from O . At this instant the distance of the other mass from O is :
- [JEE Advanced-2015]**



- (A) $\frac{2}{3}R$ (B) $\frac{1}{3}R$ (C) $\frac{3}{5}R$ (D) $\frac{4}{5}R$

- 19.** The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B , respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of n is. **[JEE Advanced-2015]**

20. A uniform wooden stick of mass 1.6 kg and length ℓ rests in an inclined manner on a smooth, vertical wall of height h ($< \ell$) such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/ℓ and the frictional force f at the bottom of the stick are: ($g = 10 \text{ ms}^{-2}$)

[JEE Advanced-2016]

- (A) $\frac{h}{\ell} = \frac{\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$ (B) $\frac{h}{\ell} = \frac{3}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$
 (C) $\frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{8\sqrt{3}}{3} \text{ N}$ (D) $\frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} \text{ N}$

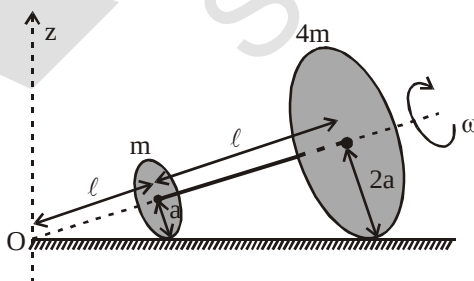
21. The position vector $\vec{r}$ of a particle of mass m is given by the following equation $\vec{r}(t) = \alpha t^3 \hat{i} + \beta t^2 \hat{j}$, where $\alpha = \frac{10}{3} \text{ ms}^{-3}$, $\beta = 5 \text{ ms}^{-2}$ and $m = 0.1 \text{ kg}$. At $t = 1 \text{ s}$, which of the following statement(s) is(are) true about the particle?

[JEE Advanced-2016]

- (A) The velocity $\vec{v}$ is given by $\vec{v} = (10\hat{i} + 10\hat{j}) \text{ ms}^{-1}$
 (B) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L} = -\left(\frac{5}{3}\right) \hat{k} \text{ Nms}$
 (C) The force $\vec{F}$ is given by $\vec{F} = (\hat{i} + 2\hat{j}) \text{ N}$
 (D) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau} = -\left(\frac{20}{3}\right) \hat{k} \text{ Nm}$

22. Two thin circular discs of mass m and $4m$, having radii of a and $2a$, respectively, are rigidly fixed by a massless, right rod of length $\ell = \sqrt{24}a$ through their center. This assembly is laid on a firm and flat surface, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω . The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure). Which of the following statement(s) is(are) true?

[JEE Advanced-2016]



- (A) The magnitude of angular momentum of the assembly about its center of mass is $17ma^2\omega/2$
 (B) The magnitude of the z -component of $\vec{L}$ is $55ma^2\omega$
 (C) The magnitude of angular momentum of center of mass of the assembly about the point O is $81ma^2\omega$
 (D) The center of mass of the assembly rotates about the z -axis with an angular speed of $\omega/5$

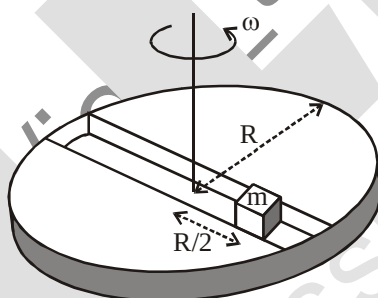
Paragraph for Question No. 23 and 24

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{\text{rot}}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{\text{in}}$ experienced by the particle in an inertial frame of reference is

$$\vec{F}_{\text{rot}} = \vec{F}_{\text{in}} + 2m(\vec{v}_{\text{rot}} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega},$$

where $\vec{v}_{\text{rot}}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x -axis along the slot, the y -axis perpendicular to the slot and the z -axis along the rotation axis ($\vec{\omega} = \omega \hat{k}$). A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at $t = 0$ and is constrained to move only along the slot. [JEE Advanced-2016]



23. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$

(B) $\frac{R}{2}\cos 2\omega t$

(C) $\frac{R}{2}\cos \omega t$

(D) $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

24. The net reaction of the disc on the block is :

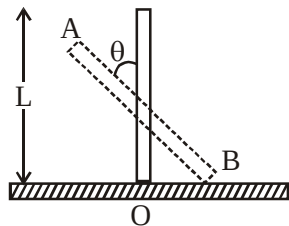
(A) $-m\omega^2 R \cos \omega t \hat{j} - mg \hat{k}$

(B) $m\omega^2 R \sin \omega t \hat{j} - mg \hat{k}$

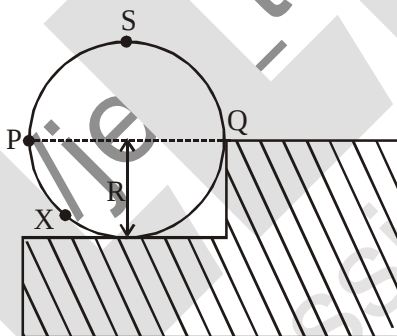
(C) $\frac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{j} + mg \hat{k}$

(D) $\frac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{j} + mg \hat{k}$

25. A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle made by the bar with the vertical is θ . Which of the following statements about its motion is/are correct ? **[JEE Advanced-2017]**



- (A) When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to $(1 - \cos\theta)$
 (B) The midpoint of the bar will fall vertically downward
 (C) Instantaneous torque about the point in contact with the floor is proportional to $\sin\theta$
 (D) The trajectory of the point A is a parabola
26. A wheel of radius R and mass M is placed at the bottom of a fixed step of height R as shown in the figure. A constant force is continuously applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q. Which of the following options is/are correct ? **[JEE Advanced-2017]**



- (A) If the force is applied normal to the circumference at point X then τ is constant
 (B) If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step
 (C) If the force is applied normal to the circumference at point P then τ is zero
 (D) If the force is applied at point P tangentially then τ decreases continuously as the wheel climbs

Paragraph for Question no. 27 & 28

One twirls a circular ring (of mass M and radius R) near the tip of one's finger as shown in Figure 1. In the process the finger never loses contact with the inner rim of the ring. The finger traces out the surface of a cone, shown by the dotted line. The radius of the path traced out by the point where the ring and the finger is in contact is r . The finger rotates with an angular velocity ω_0 . The rotating ring rolls without slipping on the outside of a smaller circle described by the point where the ring and the finger is in contact (Figure 2). The coefficient of friction between the ring and the finger is μ and the acceleration due to gravity is g .

[JEE Advanced-2017]

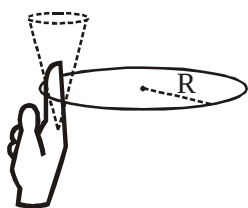


Figure 1

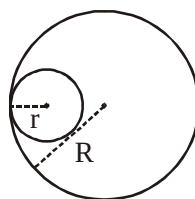


Figure 2

27. The total kinetic energy of the ring is :-

(A) $M\omega_0^2 R^2$ (B) $M\omega_0^2 (R-r)^2$ (C) $\frac{1}{2}M\omega_0^2 (R-r)^2$ (D) $\frac{3}{2}M\omega_0^2 (R-r)^2$

28. The minimum value of ω_0 below which the ring will drop down is :-

(A) $\sqrt{\frac{3g}{2\mu(R-r)}}$ (B) $\sqrt{\frac{g}{\mu(R-r)}}$ (C) $\sqrt{\frac{2g}{\mu(R-r)}}$ (D) $\sqrt{\frac{2g}{2\mu(R-r)}}$

29. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O . If v is the speed of the particle and L is the magnitude of its angular momentum about O , which of the following statements is (are) true ?

[JEE Advanced-2018]

(A) $v = \sqrt{\frac{k}{2m}}R$ (B) $v = \sqrt{\frac{k}{m}}R$ (C) $L = \sqrt{mk}R^2$ (D) $L = \sqrt{\frac{mk}{2}}R^2$

30. Consider a body of mass 1.0 kg at rest at the origin at time $t = 0$. A force $\vec{F} = (\alpha \hat{i} + \beta \hat{j})$ is applied on the body, where $\alpha = 1.0 \text{ N s}^{-1}$ and $\beta = 1.0 \text{ N}$. The torque acting on the body about the origin at time $t = 1.0 \text{ s}$ is $\vec{\tau}$. Which of the following statements is (are) true? **[JEE Advanced-2018]**

[JEE Advanced-2018]

(A) $|\vec{\tau}| = \frac{1}{3} N m$

(B) The torque $\vec{\tau}$ is in the direction of the unit vector $+\hat{k}$

(C) The velocity of the body at $t = 1$ s is $\vec{v} = \frac{1}{2}(\hat{i} + 2\hat{j})\text{ms}^{-1}$

(D) The magnitude of displacement of the body at $t = 1$ s is $\frac{1}{6}$ m

31. A ring and a disc are initially at rest, side by side, at the top of an inclined plane which makes an angle 60° with the horizontal. They start to roll without slipping at the same instant of time along the shortest path. If the time difference between their reaching the ground is $(2 - \sqrt{3})/\sqrt{10}$ s, then the height of the top of the inclined plane, in meters, is _____. Take $g = 10 \text{ ms}^{-2}$.

[JEE Advanced-2018]

ANSWER KEY

EXERCISE (S-1)

1. Ans. 2 2. Ans. $\frac{ml^2}{12}$ 3. Ans. 2.00 N·m, (b) $\hat{k}$
 4. Ans. 8 kg 5. Ans. 150(+i) 6. Ans. 30°
 7. Ans. 10^{-3} rad/s² clockwise 8. Ans. 2 9. Ans. $\frac{L}{3}$
 10. Ans. (a) $\frac{9g}{7}$ ↓ (b) $\frac{4mg}{7}$ ↑ 11. Ans. (i) 10/13m/s², (ii) 5000/26π, (iii) 480/13N
 12. Ans. 2N 13. Ans. $M = 2m \left(\frac{2gh}{R^2 \omega^2} - 1 \right)$ 14. Ans. $\sqrt{5gR}$
 15. Ans. (a) $\omega = \sqrt{\frac{6g}{\ell}}$, (b) $T = \frac{11mg}{4}$ 16. Ans. $\frac{\omega}{3}$ 17. Ans. $\frac{4\pi}{5}$
 18. Ans. 75 J 19. Ans. 1/2 ma 20. Ans. 15 21. Ans. 16 m/s²
 22. Ans. 6 mv₀² 23. Ans. $\frac{13}{16} Mv^2$
 24. Ans. $\frac{l^2}{12x}$ 25. Ans. (i) $\frac{J}{m}$ (ii) zero (iii) $\frac{J}{2m}$ (iv) $\frac{5}{2} \frac{J}{m}$
 26. Ans. (a) $t = \frac{\pi Rm}{2I}$; (b) $s = \frac{\pi R}{2}$ 27. Ans. $\frac{mv}{4}$ 28. Ans. $\frac{2mv_0^2}{l}$

EXERCISE (S-2)

1. Ans. 57/140MR² 2. Ans. $\frac{a\ell^4}{36}$ 3. Ans. (3/8) F_g 4. Ans. $a = \left[\frac{b - \mu a}{3b + \mu a} \right] g$
 5. Ans. 1.63 N, 1.224 m 6. Ans. (a) 5 m/s² →, (b) 0.3 < h < 1.5 m
 7. Ans. (a) $\sqrt{3} m/\omega^2$ (b) $(F_{\text{net}})_x = -\frac{F}{4}$, $(F_{\text{net}})_y = \sqrt{3} m/\omega^2$ 8. Ans. $\sqrt{15}$ ft/sec
 9. Ans. (a) $4\sqrt{\frac{3}{7}}$ m/s, (b) $\frac{200}{7}$ N 10. Ans. $t = 2/\sqrt{5}$ sec, $\theta = 4\pi/5$ rad
 11. Ans. (i) (a) $\frac{1.2g}{c}$ (cw) (b) $-0.3(\hat{i} + 2\hat{j})g$ (ii) (a) 24g/17c (cw) (b) 12g/17↓ (iii) (a) 2.4g/c (cw) (b) 0.5g ↓
 12. Ans. 3 13. Ans. $\alpha_{\text{max}} = \frac{2g}{3r}$
 14. Ans. (a) $a_c = \frac{4F}{3m_1 + 8m_2}$, $a_p = 2a_c$
 (b) friction at the top of the cylinder = 3m₁F/(3m₁+8m₂) towards right; friction at the bottom = m₁F/(3m₁+8m₂) towards right.

15. Ans. (a) pipe rolls without sliding (b) pipe : $5/6 \text{ m/s} \rightarrow$, $125/12 \text{ rad/s}$ (anticlockwise);
plate : $5/3 \text{ m/s} \rightarrow$

16. Ans. 6N , $-0.6\hat{j} \pm 0.6\hat{k}$ 17. Ans. (a) $\theta_c = \cos^{-1}(4/7)$, (b) $v = \sqrt{4/7 g R}$, (c) $K_T/K_R = 6$

18. Ans. $v = \sqrt{\frac{14gR}{3}}$ 19. Ans. (a) 6.68 m/s (b) 2.27 m

20. Ans. (i) $2v_0/3$, (ii) $t = v_0/3\mu g$, $W = \frac{1}{2}[3\mu^2 m g^2 t^2 - 2\mu m g t v_0]$ ($t < t_0$), $W = -\frac{1}{6}mv_0^2$ ($t > t_0$)

21. Ans. (a) $\omega/3$, (b) $\frac{\sqrt{37}}{3}m\omega R$, (c) $\frac{\sqrt{37}}{3}m\omega R$ 22. Ans. 60°

23. Ans. (a) $\ell = 0.1\text{m}$; (b) $\omega' = 1\text{rad/s}$; (c) laminar sheet will never come to rest

24. Ans. 5 25. Ans. (a) $\frac{m}{M} = \frac{1}{4}$; (b) $x = \frac{2L}{3}$; (c) $\frac{v_0}{2\sqrt{2}}$

EXERCISE (O-1)

- | | | | | | |
|---|--------------------|----------------|----------------|------------------|--------------|
| 1. Ans. (C) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (D) | 5. Ans. (A) | 6. Ans. (A) |
| 7. Ans. (B) | 8. Ans. (D) | 9. Ans. (C) | 10. Ans. (A) | 11. Ans. (C) | 12. Ans. (B) |
| 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (B) | 16. Ans. (C) | 17. Ans. (B) | 18. Ans. (D) |
| 19. Ans. (C) | 20. Ans. (C) | 21. Ans. (A) | 22. Ans. (B) | 23. Ans. (C) | 24. Ans. (D) |
| 25. Ans. (A) | 26. Ans. (D) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. (A) | 30. Ans. (D) |
| 31. Ans. (B) | 32. Ans. (B) | 33. Ans. (A) | 34. Ans. (A) | 35. Ans. (B) | 36. Ans. (C) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (A) | 40. Ans. (D) | 41. Ans. (B) | 42. Ans. (D) |
| 43. Ans. (C) | 44. Ans. (A) | 45. Ans. (C) | 46. Ans. (D) | 47. Ans. (B) | 48. Ans. (B) |
| 49. Ans. (A) | 50. Ans. (A,B,C,D) | 51. Ans. (B,C) | 52. Ans. (A,D) | 53. Ans. (A,C,D) | |
| 54. Ans. (A,C,D) | 55. Ans. (A,B,C,D) | 56. Ans. (A) | 57. Ans. (A) | | |
| 58. Ans. (D) | 59. Ans. (B) | 60. Ans. (A) | 61. Ans. (D) | 62. Ans. (D) | |
| 63. Ans. (A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (T); (D) $\rightarrow$ (Q,S) | | | | | |

EXERCISE (O-2)

- | | | | | | |
|--|------------------|------------------|--------------------|------------------|--------------|
| 1. Ans. (D) | 2. Ans. (C) | 3. Ans. (B) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (D) |
| 7. Ans. (B) | 8. Ans. (B) | 9. Ans. (B) | 10. Ans. (B) | 11. Ans. (D) | 12. Ans. (C) |
| 13. Ans. (A) | 14. Ans. (C) | 15. Ans. (C) | 16. Ans. (D) | 17. Ans. (C) | 18. Ans. (A) |
| 19. Ans. (C) | 20. Ans. (C,D) | 21. Ans. (B,C,D) | 22. Ans. (A,B,C) | 23. Ans. (A,B,D) | |
| 24. Ans. (A,C) | 25. Ans. (A,B,C) | 26. Ans. (A,B,C) | 27. Ans. (B,C,D) | | |
| 28. Ans. (A,C,D) | 29. Ans. (A,C) | 30. Ans. (B,D) | 31. Ans. (A, B, C) | | |
| 32. Ans. (C) | 33. Ans. (A) | 34. Ans. (C) | 35. Ans. (C) | 36. Ans. (B) | 37. Ans. (C) |
| 38. Ans. (B) | 39. Ans. (B) | 40. Ans. (C) | 41. Ans. (C) | 42. Ans. (B) | 43. Ans. (B) |
| 44. Ans. (C) | 45. Ans. (B) | 46. Ans. (A) | 47. Ans. (C) | 48. Ans. (A,B,C) | |
| 49. Ans. (C) | 50. Ans. (B,C,D) | 51. Ans. (A,C,D) | | | |
| 52. Ans. (A) ST (B) R (C) PRT (D) PQ 53. Ans. (A)-R, (B)-S, (C)-P, (D)-Q | | | | | |

EXERCISE (JM)

- | | | | | | |
|--------------|--------------|--------------|-------------------|--------------|--------------|
| 1. Ans. (2) | 2. Ans. (4) | 3. Ans. (4) | 4. Ans. (2) | 5. Ans. (4) | 6. Ans. (3) |
| 7. Ans. (1) | 8. Ans. (4) | 9. Ans. (1) | 10. Ans. (1 or 3) | 11. Ans. (2) | 12. Ans. (3) |
| 13. Ans. (3) | 14. Ans. (4) | 15. Ans. (3) | | | |

EXERCISE (JA)

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|--------------|------------------------|--------------------|----------------------------|--------------|-------------------|
| 1. Ans. (B) | 2. Ans. (A) | 3. Ans. (B,C) | 4. Ans. 4 | 5. Ans. 9 | 6. Ans. (AC or C) |
| 7. Ans. (B) | 8. Ans. (D) | 9. Ans. 3 | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (A,B) |
| 13. Ans. (D) | 14. Ans. 8 | 15. Ans. 4 | 16. Ans. 2 | 17. Ans. 7 | 18. Ans. (C, D) |
| 19. Ans. 6 | 20. Ans. (D) | 21. Ans. (A, B, D) | | 22. Ans. (D) | 23. Ans. (D) |
| 24. Ans. (C) | 25. Ans. (A), (B), (C) | | 26. Ans. (B,C or C) | | 27. Ans. (Bonus) |
| 28. Ans. (B) | 29. Ans. (B,C) | 30. Ans. (A,C) | 31. Ans. 0.75 [0.74, 0.76] | | |